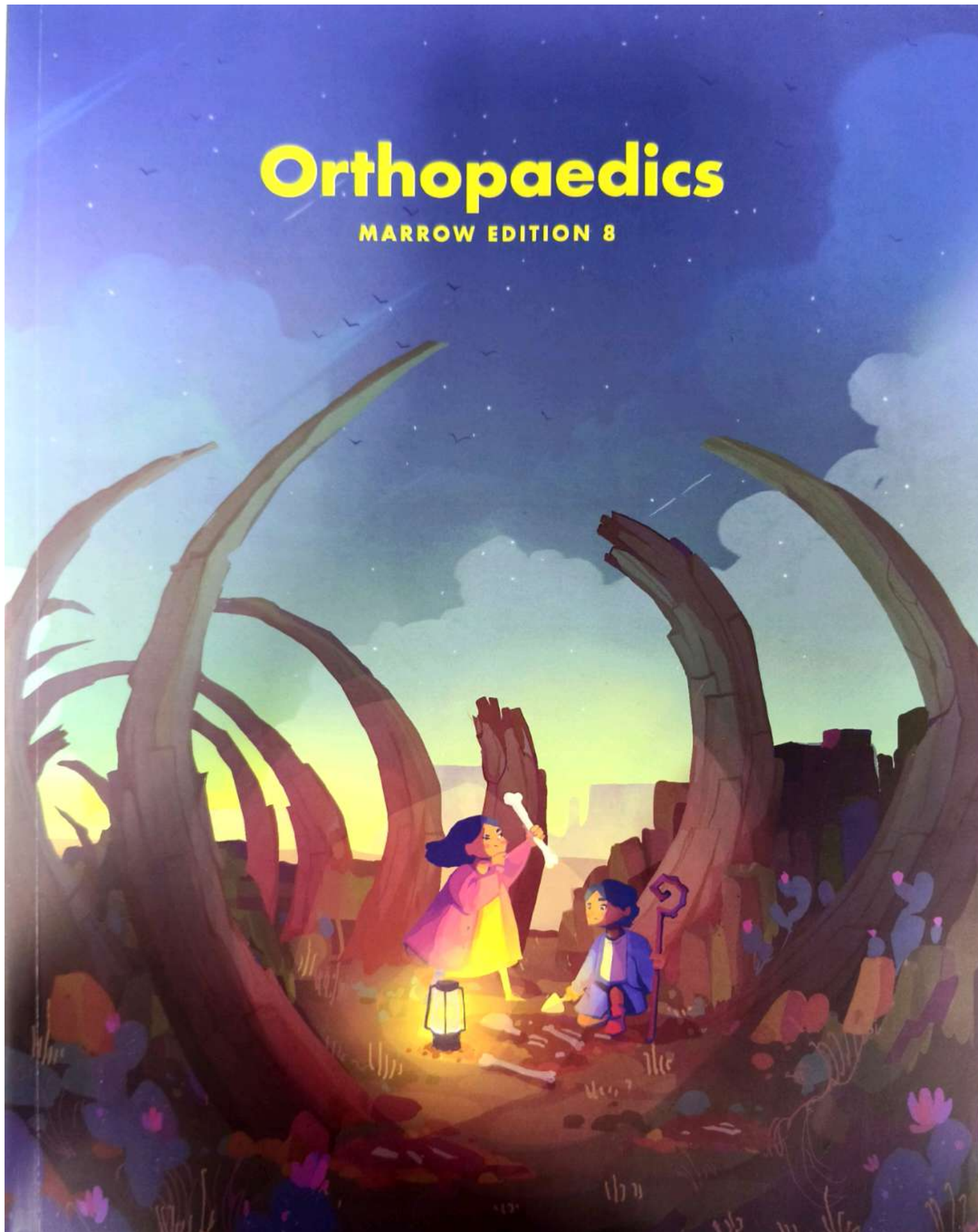


Orthopaedics

MARROW EDITION 8



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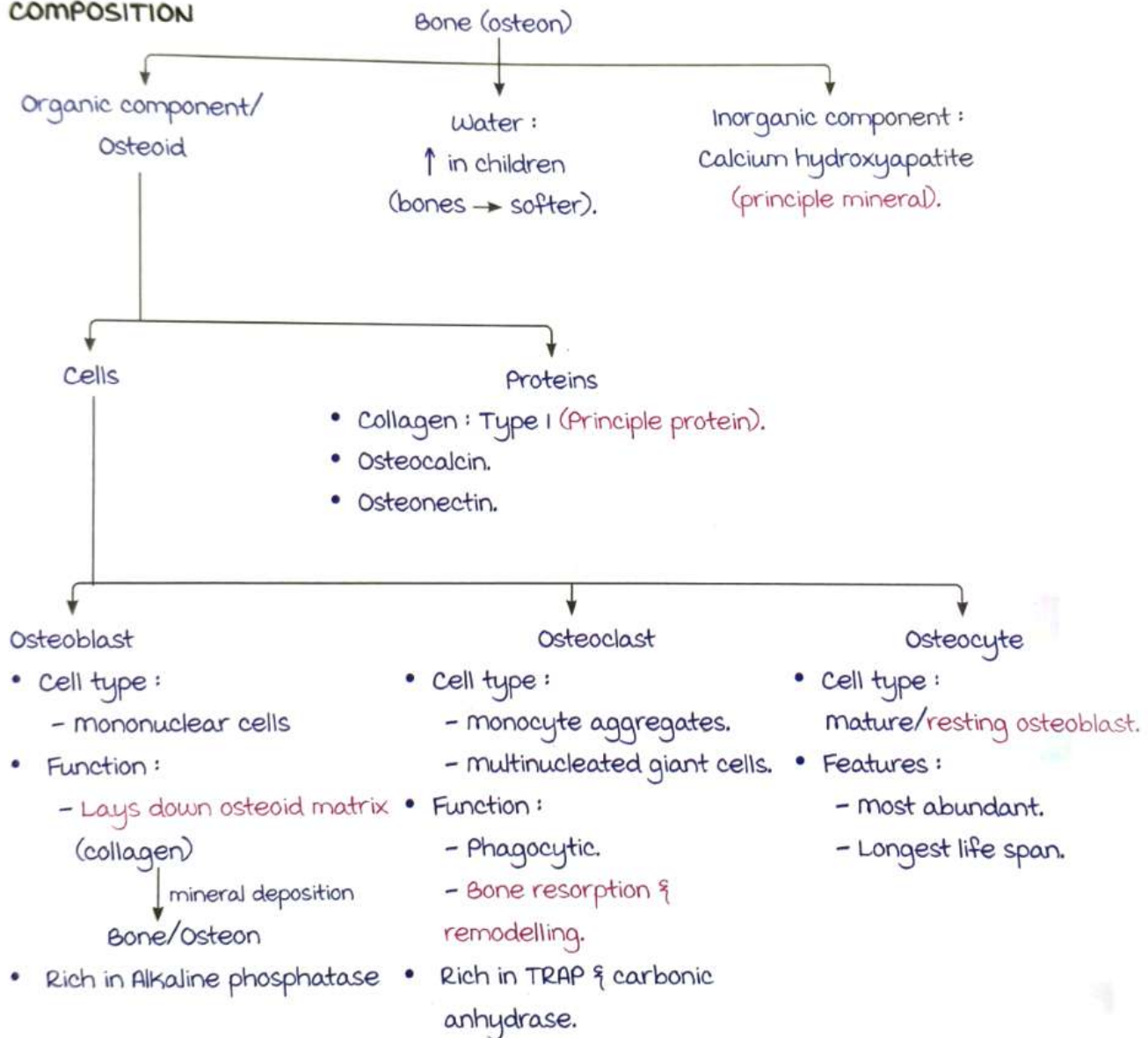
BASIC ANATOMY AND PHYSIOLOGY OF BONES

----- Active space -----

Bone composition & markers

00:00:38

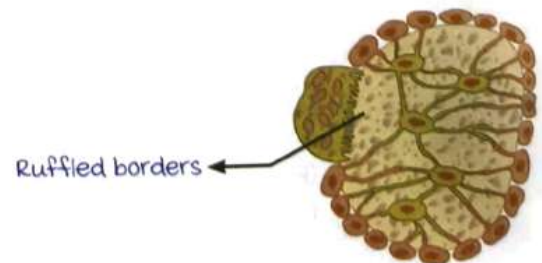
COMPOSITION



Note : Collagen types

Type 1	Type 2
- Bones. - meniscus. - Annulus fibrosis.	- Articular hyaline cartilage. - Nucleus pulposus.

- Strength of bone depends on osteoblasts & osteoclasts.



----- Active space -----

BONE MARKERS

markers

Bone formation

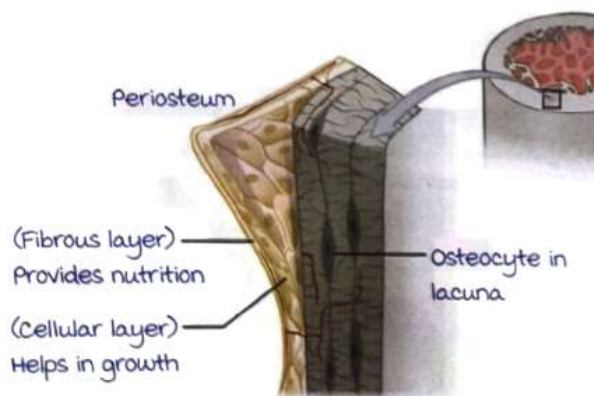
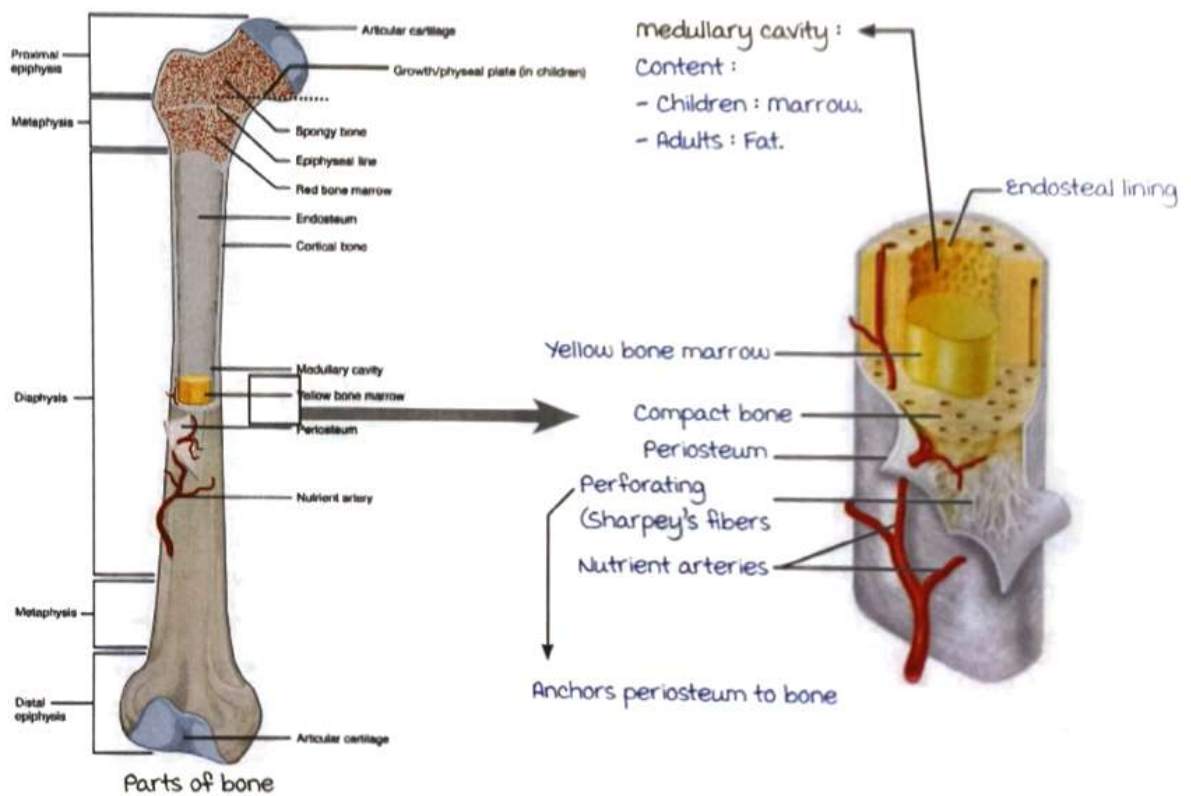
- Alkaline phosphatase (ALP)
- Procollagen I
- Osteocalcin
- Osteonectin

Bone resorption

- Hydroxyproline } Collagen breakdown
- Hydroxylysine } products
- N⁹C telopeptide
- TRAP

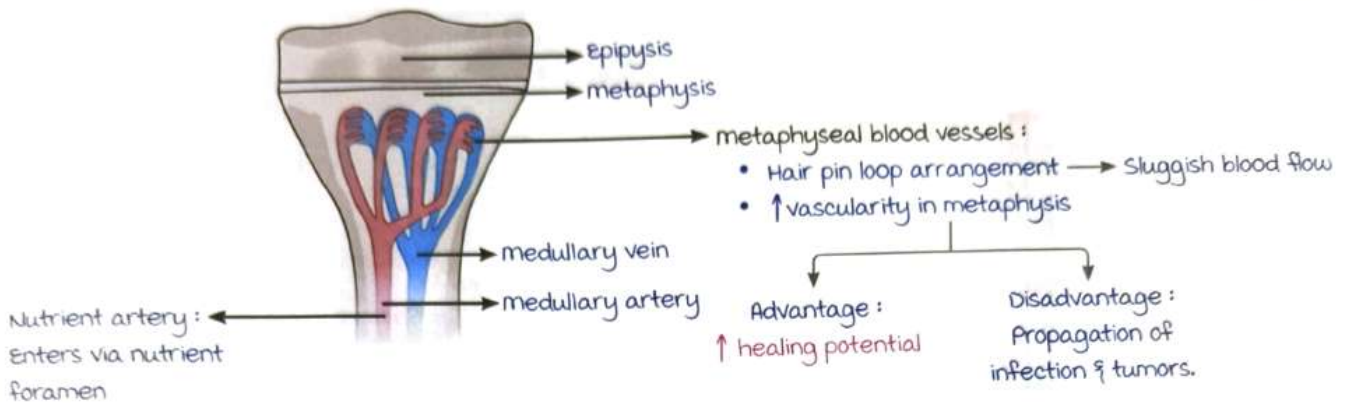
Anatomy

00:09:28

**Periosteum :**

- Present in all long bones
- Absent in :
 - Articular surface of long bones.
 - Sesamoid bones (eg : Patella).

----- Active space -----



Note : metaphysis has very few macrophages/monocytes.

TYPES OF BONE ARCHITECTURE

Immature woven bone :

- Improperly arranged collagen.
- Weak.
- Seen in :
 - Children.
 - Bones healing from fracture (eg : callus).

mature/lamellar bone :

	Cancellous/spongy	Cortical/compact
Site	metaphysis	Diaphysis
Consistency	Soft	Hard
Bone cells	↓	↑
vascularity	↑	↓
Fracture healing	Fast	Slow
Risk of infection	↑	↓
Fracture complication	malunion	Non union

Note :

Epiphyseal fractures (Intraarticular) → Osteoarthritis → ↓ Range of motion.

Bone Growth

00:20:40



a). Lateral view

b). Anteroposterior view

X-ray : Left hand (child).

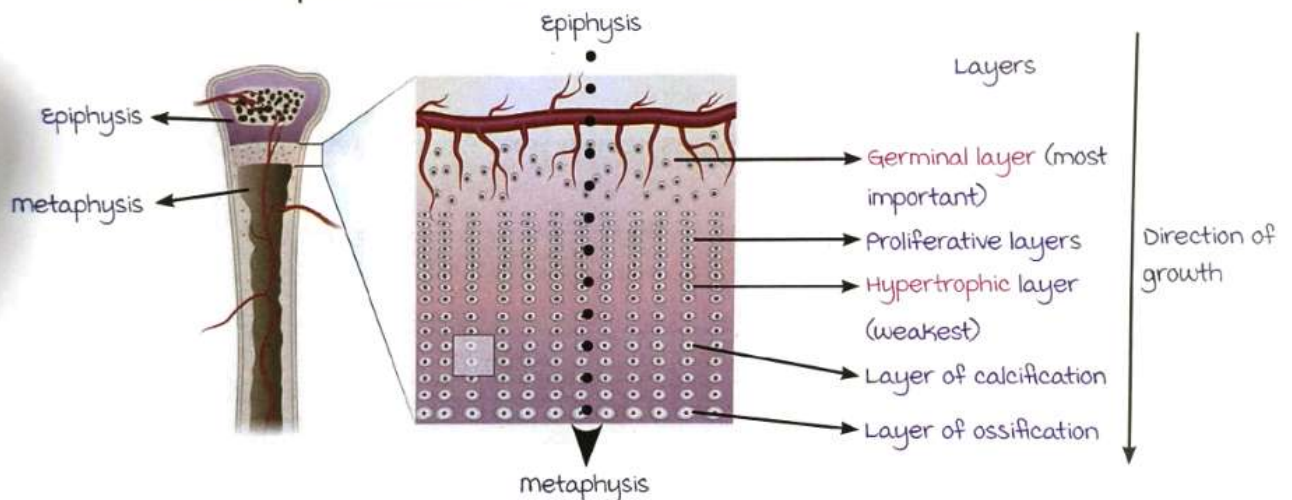
Epiphysis

Physis : Only in children

↑ ses length of bone

metaphysis

Growth plate structure :



Applied aspect :

Growth plate (germinal layer) injury

Complete

Partial

- I/L side (bone) : Stops growing.
- C/L side (bone) : Continues to grow.

Limb length discrepancy

Asymmetrical growth

Deformity (varus/valgus)

Appositional growth (↑ ses thickness) : By endosteum & periosteum.

BASIC TERMINOLOGY OF FRACTURE AND HEALING

----- Active space -----

Fractures

00:00:49

Break in continuity of bone cortex.

GENERAL FEATURES OF FRACTURE

Clinical signs :

- Abnormal mobility at fracture site > crepitus (Pathognomonic).
- Tenderness (most consistent).
- Swelling.

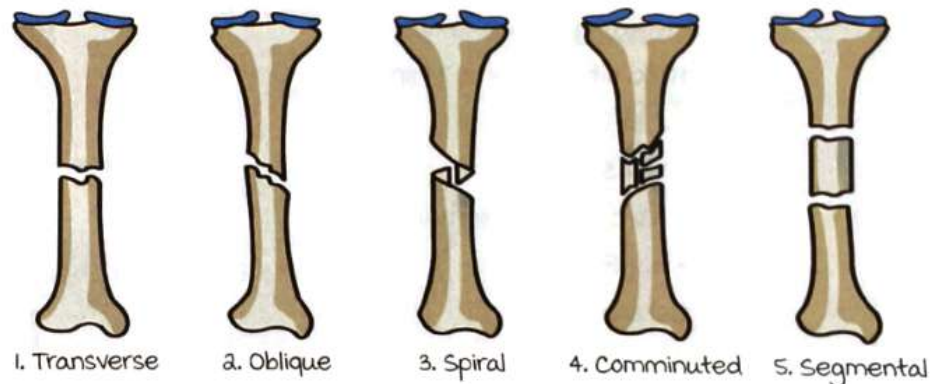
Radiological :

- X-ray : Discontinuity of bone.

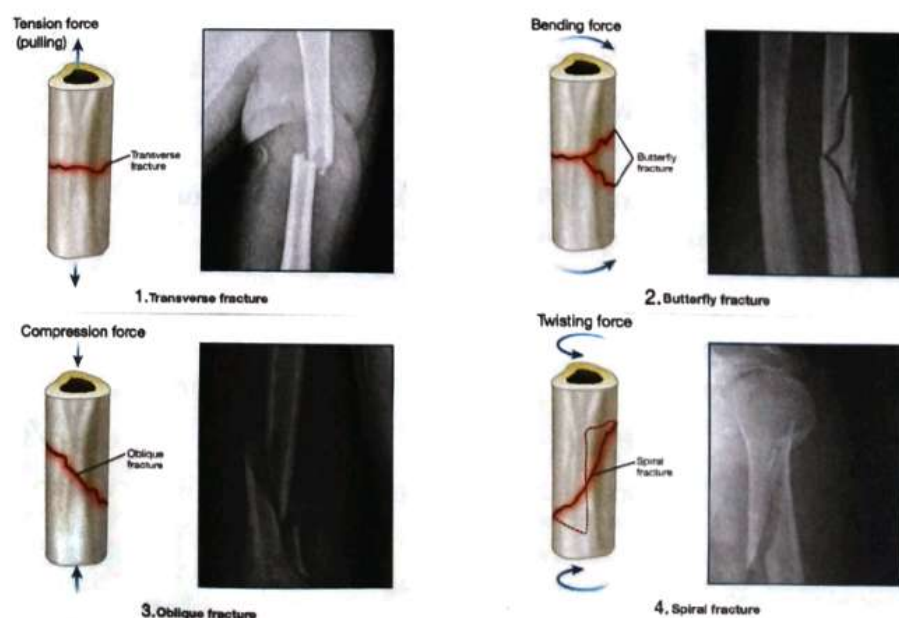
CAUSES

Fractures with significant trauma :

Types :



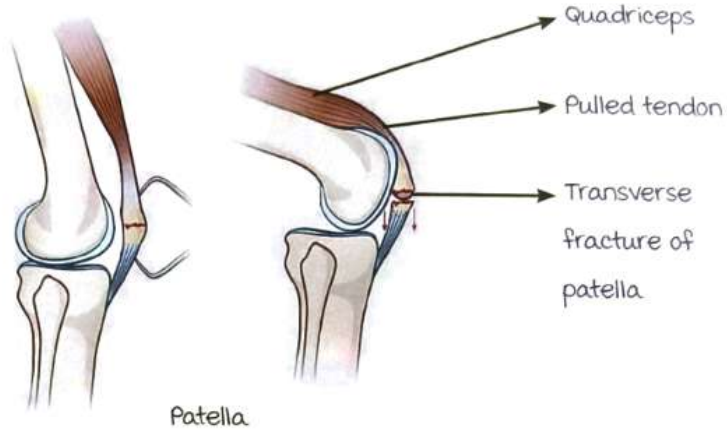
mechanism of injury :



----- Active space -----

Avulsion fractures :

- Violent pull of tendons → Avulsion of bone.
- Eg : Patella, olecranon.



Fractures without significant trauma : Pathological fractures

Localized causes

- Infection.
- Ischemia.
- Lesions.
- Cysts.
- Radiation.

Generalized causes

- Osteoporosis (m/c cause) :
m/c sites : spine > hip > wrist (Colle's).
- metastasis :
- m/c site : Proximal femur.
- Ix : FDG-PET
- Osteomalacia.
- Rickets.
- Osteogenesis imperfecta.
- Paget's disease.
- Scurvy.

- Pain before the fracture (d/t micro fractures).

Investigation :

- X-ray : Shows primary lesion.

management :

A. mirel's criteria :

- used to predict risk of pathological fracture d/t lesions.
- Score of ≥ 8 : Requires prophylactic fixation with internal fixation devices.

B. Treat underlying cause.



Fracture 2° to cyst



Fracture 2° to Paget's disease

Stress fracture :

- D/t sudden & repetitive loading of bone.
- Common sites :
 - a. Tibia.
 - b. Fibula.
 - c. metatarsal (march fracture) : } weight bearing bones
 - 2nd > 3rd metatarsal.
 - Neck > shaft.
- Pain :
 - Prior to fracture.
 - ↑ on activity.
- Investigations :
 - X-ray : Fracture visible after 2-3 wks of symptoms.
 - MRI (IOC) :
 - Identifies → Soft tissue edema
 - Occult fractures.
 - Bone scan : For multiple stress fractures.
- management :
 - Stop the stressors.
 - Immobilization of affected limb.



march fracture



Runner's fracture

Shin splints :

AKA medial Tibial Stress Syndrome (MTSS).

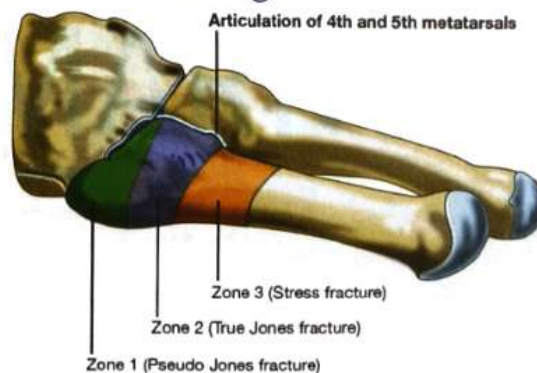
- Seen in runners.
- medial periostitis of tibia.
- mechanism of injury :
 - Repetitive muscle contractions → inflammation of periosteum at site of muscle attachment

↓
Pain over anteromedial side of leg

- X-ray : Normal.

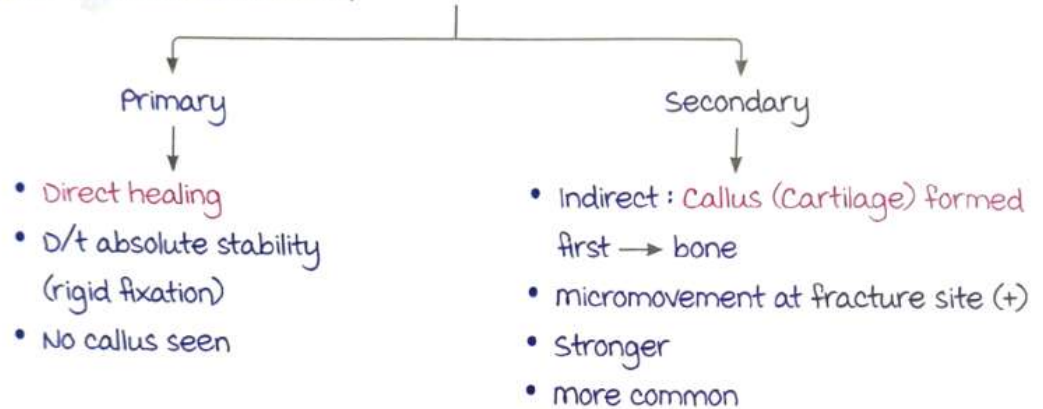
Note :

Robert Jones : Fracture of 5th metatarsal base.



Robert Jones fracture

TYPES OF FRACTURE HEALING



1 Hematoma formation (2-3 days)



2 Granulation Tissue Formation (Inflammation) (2-3 weeks)



3 Callus Formation (2-3 months)



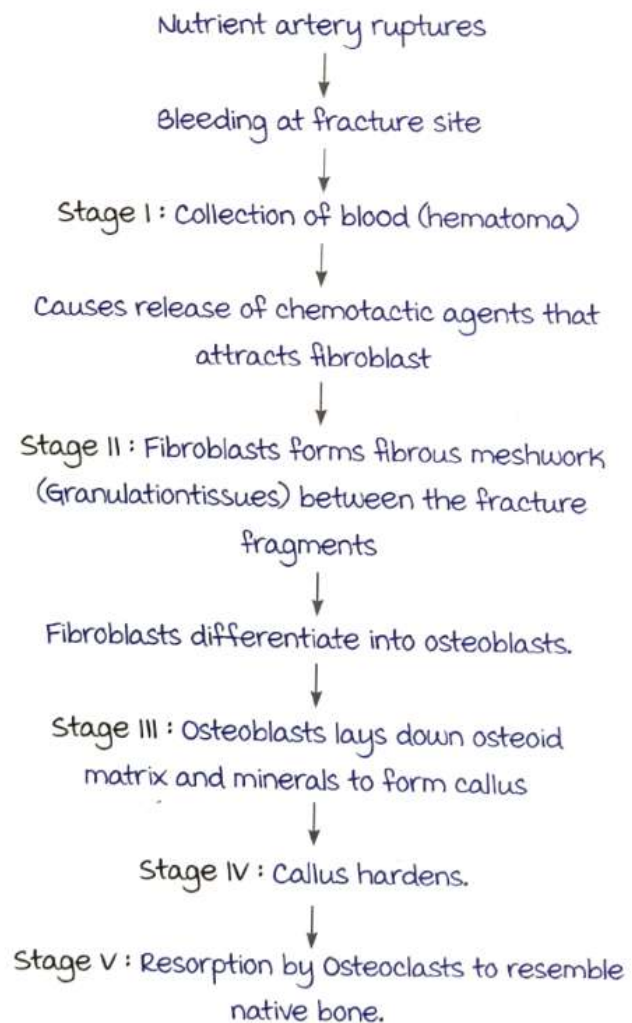
4 Consolidation (2-3 years)



5 Bone remodelling (3 years)



STAGES OF FRACTURE HEALING (2°)



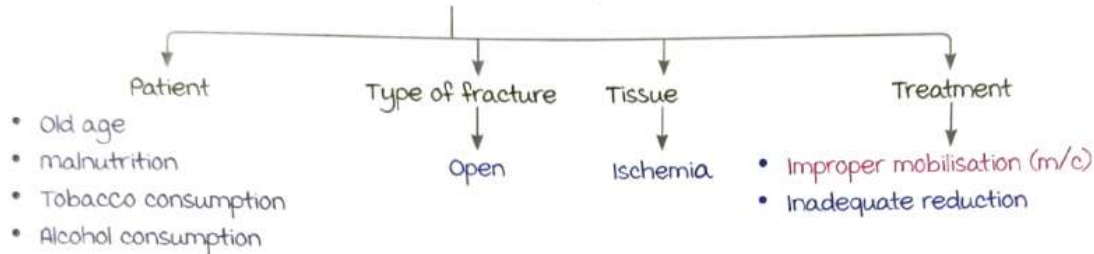
Callus :

- Disorganised mass of immature bone.
- **visible on x-ray** after 2-3 weeks of fracture.

----- Active space -----

Note : compression at fracture site → promotes union.

FACTORS INHIBITING FRACTURE HEALING



Complications in fracture healing

00:26:19

NON-UNION

- Callus formation absent for > 9 months.
- Risk factor : Diaphyseal bone fractures.
- m/c site : **Lower 1/3rd Tibia**.

Types :



Differences between Hypertrophic & Atrophic Non-Union :

	Hypertrophic	Atrophic (typical)
Ends of fracture	Smooth & sclerosed (↑ density)	
Fracture line	Visible	
x-ray appearance		
Callus formation	↑↑	Absent
Bone biology	Good	Abnormal
Immobilisation	Improper	±
Treatment	Immobilization	Autologous bone grafting

----- Active space -----

Bone Graft :

m/c source : Iliac crest.

Ideal bone graft characteristics :

1. Osteogenesis : Graft itself makes bone using osteoblasts.
2. Osteoconduction : Allows bone to grow on it as a scaffold.
3. Osteoinduction : Stimulates surrounding host bone to form bone
(Growth factors : differentiates mesenchymal cells → bone forming cells).

Substitutes of bone graft :

Property	Substitute
Bone conduction	• Calcium phosphate • Calcium sulphate • PMMA (Poly methyl methacrylate)
Bone induction	Bone morphogenic protein (improves bone union)

MALUNION

- Fractures healing in anatomically abnormal position.
- Rx : Osteotomy (Bone realigned & fixed if functional abnormality (+)).

Note : Bones with ↑ risk of malunion vs non-union.

malunion	Non-union
• Clavicle (m/c) • Supra-condylar humerus • Colles • Intertrochanteric (extracapsular) femur	• Lower 1/3rd of tibia (m/c) • Scaphoid • Lateral condyle of humerus • Neck of femur (intracapsular) • Neck of Talus

Fractures that undergo malunion, rarely/never undergo non-union (& vice-versa).

Management of Fracture

00:36:42

Important Terms :

Displacement : movement of distal fracture segment d/t pulling forces by muscle.

Reduction : Bringing fracture segment back to normal anatomy with counter-forces.

Traction : Counter-force applied.

Fixation : maintaining fracture in reduced position till it heals.

REDUCTION

methods :



manual closed Reduction :

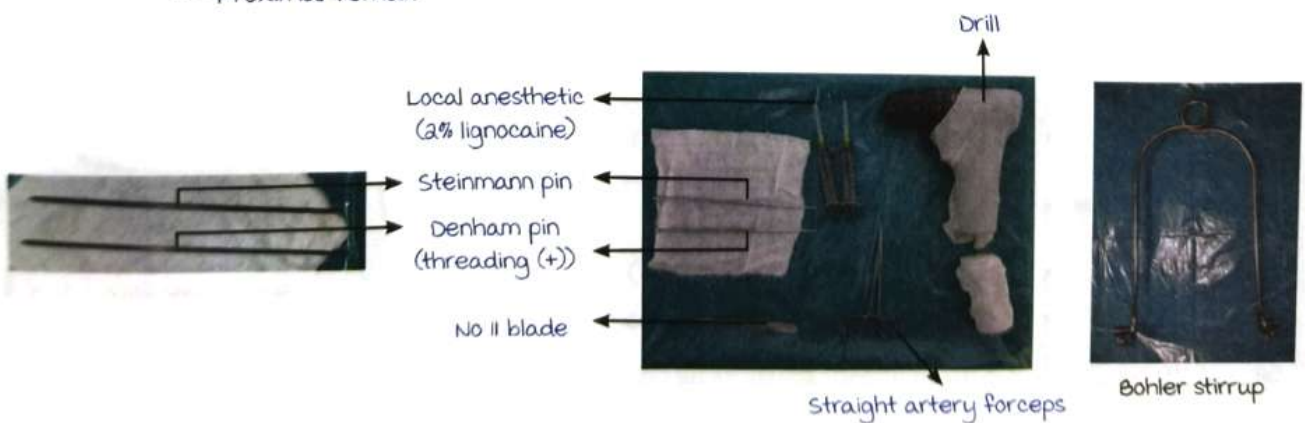
- Done using manual 'traction' and counter traction.
- Followed by POP application.
- Repeat X-ray to confirm reduction.

Skin traction :

Bandage wrapped → Force on bandage → Force on skin → Force on soft tissue
→ maintains fracture reduction

Skeletal traction :

Site of pin → upper tibia (m/c).
→ Lower tibia.
→ distal femur.
→ Proximal femur.



Instruments used for skeletal traction

----- Active space -----

		Denham pin	Steinmann pin
Threading		+	-
Indications	1. Type of bone fractured	Osteoporotic	Normal
	2. Site of traction	Calcaneal	Tibia / femoral

Procedure (upper tibial skin traction) :

1. Site : Draped and prepared.
2. Local anesthetic given : Bone deep on either side of bone laterally.
3. Stab wound $\begin{cases} \rightarrow 2.5 \text{ cm below and behind tibial tubercle.} \\ \rightarrow \text{made using No. 11 scalpel reaching bone} \\ \quad \text{(to avoid friction while inserting pin).} \end{cases}$
4. Dilate entry site using artery forceps.
5. Pin placed in drill and inserted into entry wound.
6. Pin drilled through the bone and brought out at opposite site.
7. Drill removed and wound dressed.
8. Böhler's stirrup mounted onto pin and locked.
9. Weights applied using ropes.
10. Patient mounted onto a splint.

Complications :

- *Staphylococcus aureus* infection : m/c infection at pin traction sites.
- Pin-tract sequestrum : ring shaped sequestrum at entry & exit sites.

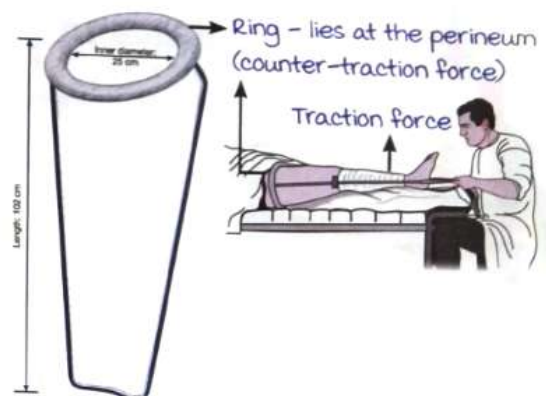
uses : for traction of lower limb fractures : Intertrochanteric, Distal femur.

TEMPORARY MANAGEMENT : SPLINTS

To stabilize fracture.

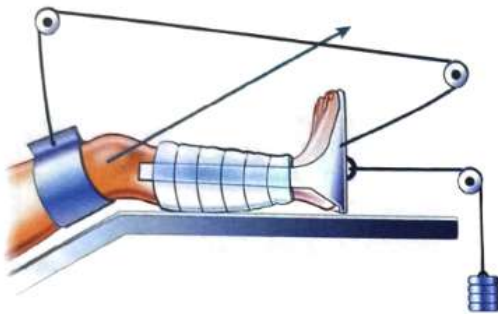
Thomas splint :

- Inventor : H.O Thomas.
- First use : to treat TB knee.
- Fixed traction device (has both traction & counter-traction forces).
- Advantage : Patient mobility (+).
- Indication : lower limb fractures (m/c : femur fractures).



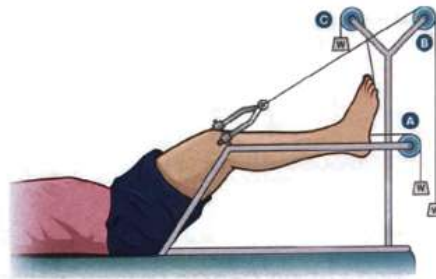
Bohler & Braun Splint :

- Sliding traction device used in wards.
- Force :
 - Traction : Weights
 - Counter traction : Gravity (footend of bed raised)
- Location of pulleys :
 - A : For Tibia fractures.
 - B : For Femur fractures.
 - C : To prevent foot drop.



Russell's Traction

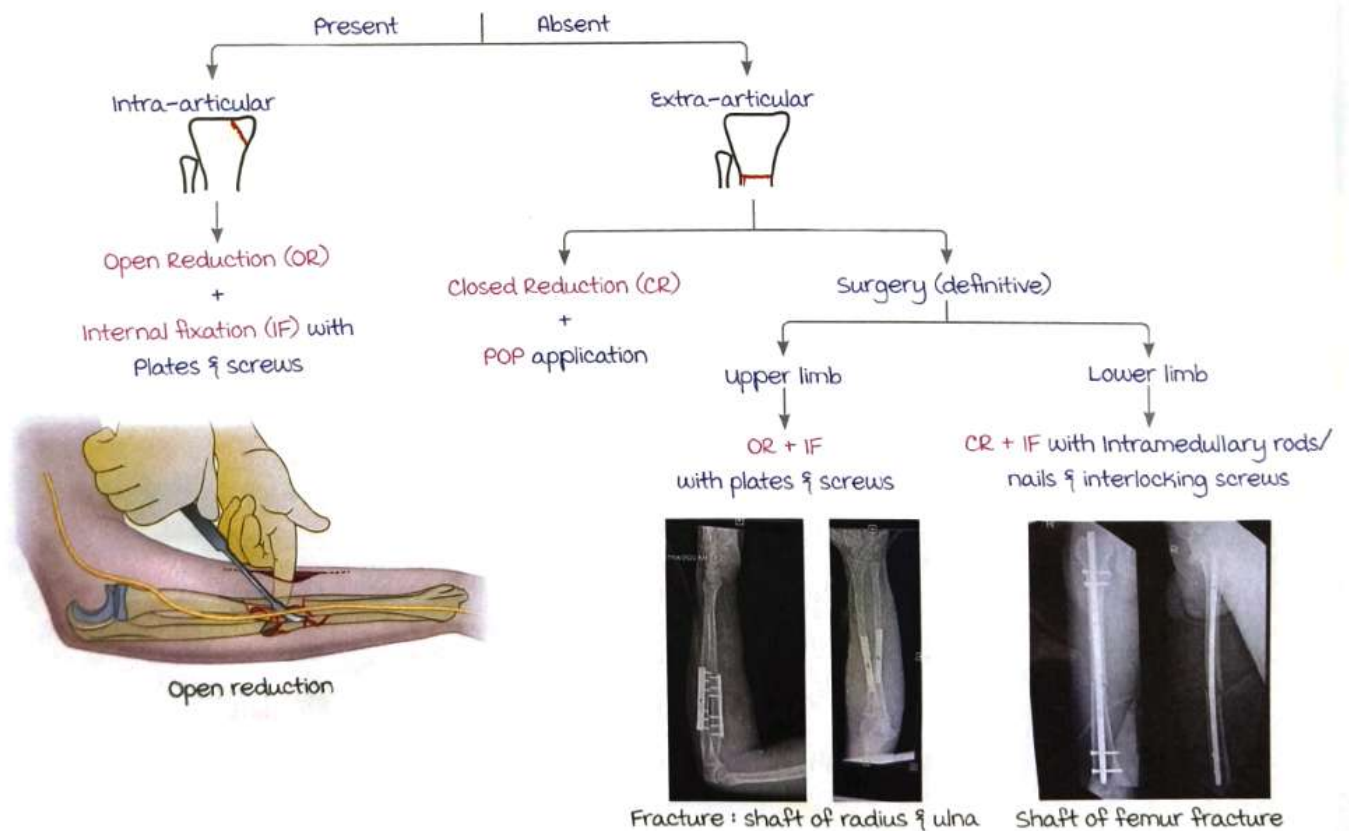
Indication : Inter-trochanteric fracture of femur



Bohler & Braun splint : location of pulleys

DEFINITIVE TREATMENT

Algorithm : Assess for involvement of articular surface.



----- Active space -----

Reduction $\begin{cases} \text{Open : Fracture visualised and reduced} \\ \text{Closed : Done without opening fracture site and confirmed with X-ray} \end{cases}$

Accuracy of reduction : $\text{Open} > \text{closed}$

Hence preferred in intra-articular fractures (to prevent osteoarthritis)

POP application :

Disadvantages	Advantages
• Cumbersome • Requires joint immobilization above & below fracture	• Non-invasive • Cheap • Easy application & removal
Disuse atrophy Stiffness	

CR+IF :

Reduction : using C-arm (image intensifier)

Fixation : Rod/nail passed into the bone medullary cavity

interlocking screws placed (to avoid rotation of rod)

Fractures due to avulsion forces :

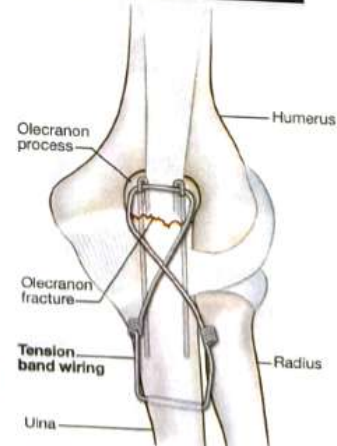
Patella fracture
(d/t Quadriceps pull)



Olecranon fracture
(d/t Triceps pull)



Rx : OR + IF with Tension Band Wiring (TBW)



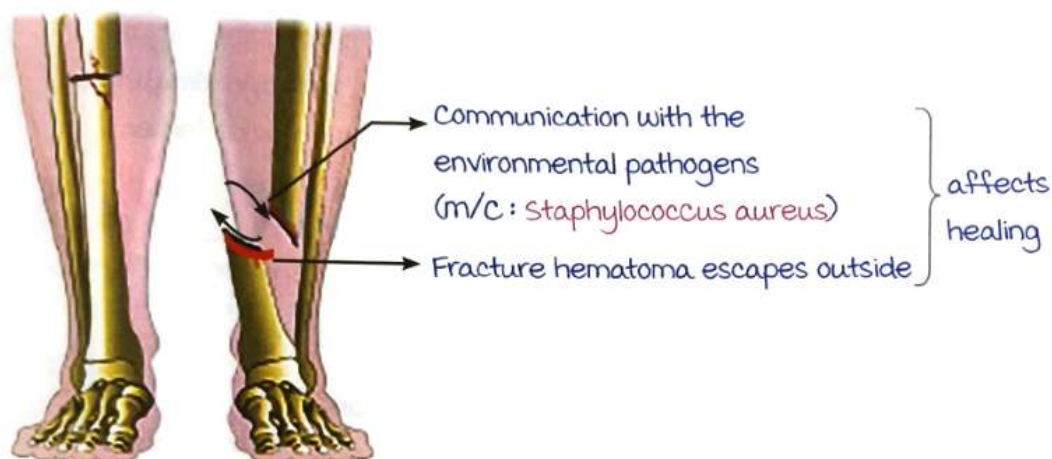
OPEN FRACTURE, AMPUTATIONS AND POLYTRAUMA

----- Active space -----

Open fracture

00:00:27

- Fracture + break in skin and underlying soft tissue.
- m/c involved : Tibia and phalanges.



GUSTILO ANDERSON CLASSIFICATION

Type I	Type II	Type IIIa	Type IIIb	Type IIIc
• Wound <1cm long. • Cause : D/t impingement of fracture fragments on skin. • AKA inside out injury.	Wound 1-10 cm	Open fracture + contaminated environment : sewage, farms or firearm injury.	Open fracture with periosteal stripping	Open fracture + vascular injury
Usually > 10 cm				

----- Active space -----

MANAGEMENT**Wound management (more important) :**

1. Antibiotics

2. Wound toilette :

- Wound wash : Povidone iodine & sterile NS.
- Wound debridement.

3. Wound closure :

Primary closure :

Only in clean wounds where the edges are approximating

+

Satisfactory debridement

Secondary/delayed closure :

- If > 6 hrs - 12 hrs (golden period).
- Associated neurovascular injury.
- Edges cannot be approximated.
- Not satisfied with debridement.

4. Wound healing : manage like a closed fracture.

Fracture :

Grade I : managed like closed fracture.

Grade II, IIIa :

- If clean (<6-12 hrs) → manage like a closed fracture.
- Otherwise like open (external fixation).

Grade IIIb, IIIc : External fixation (immobilization/stabilization of bone outside the body).

Note : Internal fixation is C/I in open fractures.

Placing a device inside the bone → growth of organisms & biofilm formation → Infection.

External Fixation

00:07:47

Fixation of bone outside the body.



Schanz pins

- Placed proximal and distal to the wound
- They immobilize the bone

Rods to immobilize the pins



Schanz pins



Connecting Rod



Clamps

stability :

↑ stability by :

- ↑ number of pins.
- ↑ number of rods.
- ↑ number of planes used to stabilize the fracture (Biplanar > uniplanar).

----- Active space -----

TYPES**1. Rail fixator/Limb reconstruction system :****2. Ilizarov ring fixator :**

- multiplanar
- **most stable external fixator.**
- Allows for compression and distraction of fracture site.
- Pins can be put in any direction.



multiplanar system

Distraction Osteogenesis :

Done in open fractures with bone loss.

Indications :

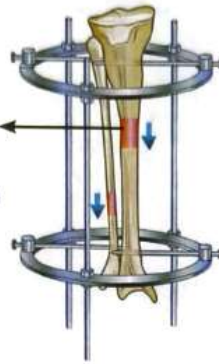
- Open fractures.
- Non union.
- Infected non union.
- Deformity correction or malunion.
- Fracture with bone loss.
- Limb lengthening.

----- Active space -----

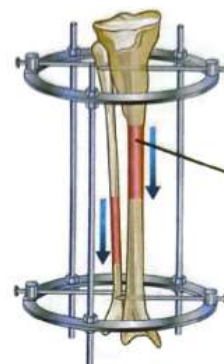


1. Corticotomy

Stretching of
callus and new
bone formation
(Callotaxis)



2. Distract the rods (1 mm/day) - physiological limit

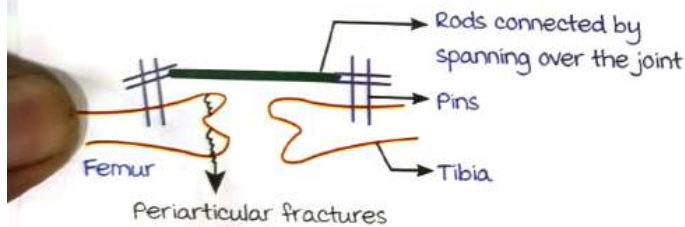


3. growth of new bone

3. Spanning external fixator :

Uses : Periarticular open fractures.

Eg : Side swipe injury/baby car injury.



Crushing of distal
humerus and
proximal ulna



Spanning external fixator

Complications :

Necrosis of site where pin is placed (infection/heat d/t drilling).

Note : Compartment syndrome.

- Equal incidence in high and low energy injuries.
- Can occur in open fractures.
- High index of suspicion and be vigilant in patients with altered level of consciousness.



Amputation

00:19:50

Damaged limb/contaminated/not salvageable

Amputation :
Cutting the limb off the body
through the bone

Disarticulation :
Cutting the limb off the body
through the joint

Indications : Dead, dangerous or dumb limbs

- Crush injury/RTA.
- Peripheral vascular disease (PVD) like diabetes mellitus.
- Frost bite, gas gangrene, malignant tumour.
- Congenital absence of limb/neuromuscular disease like polio.

----- Active space -----

Trauma scores :

1. Mangled extremity severity score (MESS) : Score ≥ 7 = Amputation.

- V : Velocity of injury or soft tissue coverage.
- I : Ischemia time (most important).
- S : Shock.
- A : Age of patient.

2. Limb salvage score

3. Ganga score

Types :

1. Closed amputation :

Skin is closed in 1^o surgery.



2. Open or guillotine :

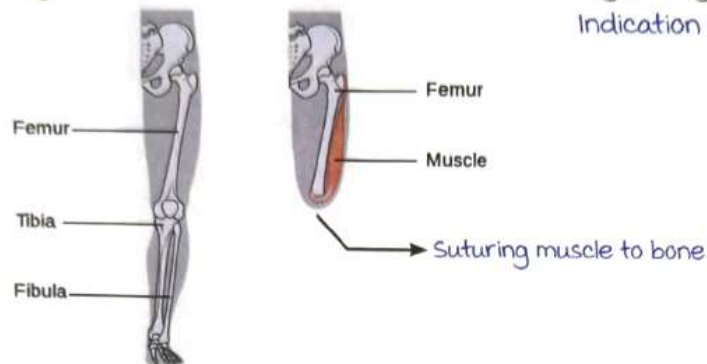
- Skin is kept open after amputation & closed in 2nd stage.
- Indications :
 - Infected/contaminated stump.
 - Amputation for ischemic conditions.



Principles :

1. Use of tourniquet : Reduces bleeding
C/I if cause of amputation is ischemia or PVD.
2. Structures cut :
 - Amputation : Everything is cut.
 - Disarticulation : Sparing of bone.
3. Bone has to be cut 5cm shorter than soft tissue.
4. Double ligate the vessels.

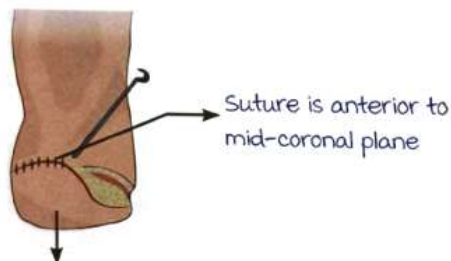
----- Active space -----

methods of closure :**1. myodesis :**

- Provides stable stump.
- C/I in PVD amputations.

2. myoplasty : Suturing muscle to muscle

Indication : Ischemia/PVD.

Skin flaps :**1. Skewed flap :**

Posterior flap is longer than anterior

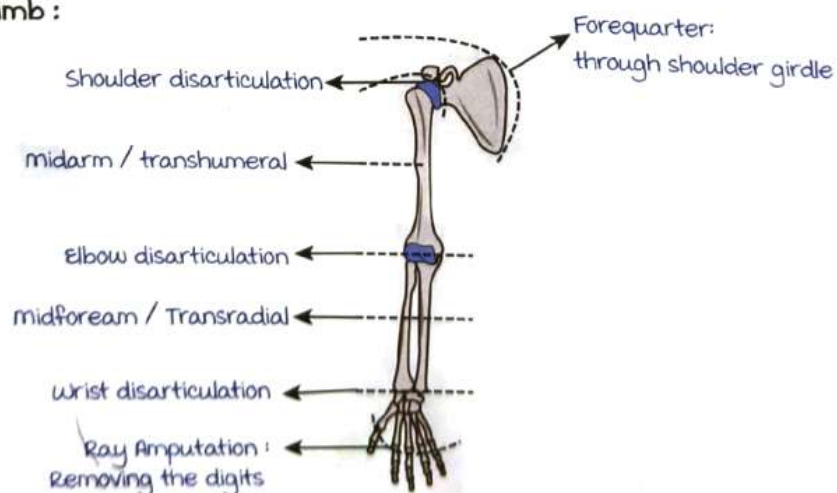
- ↓ Problems of application of the limb prosthesis and ↓ risk of wound dehiscence.

2. Symmetric flap :

- Anterior and posterior flap are of same length.
- Suture line is on the mid-coronal plane.
- Disadvantage : Pain while placing prosthesis.

Terminologies of Amputation

00 :26 :13

upper limb :

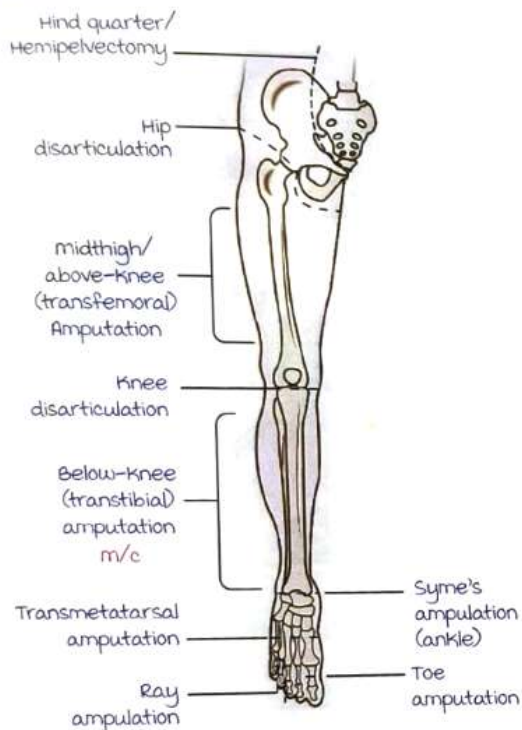
Krukenberg amputation :

----- Active space -----

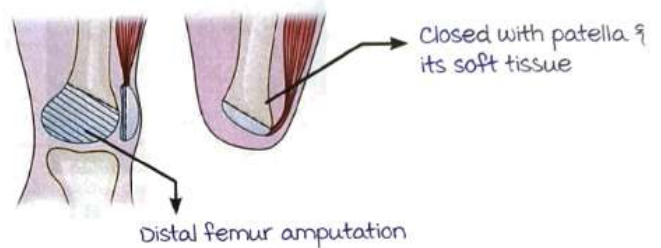


To use radius & ulna
as pincers - performed
in blind people

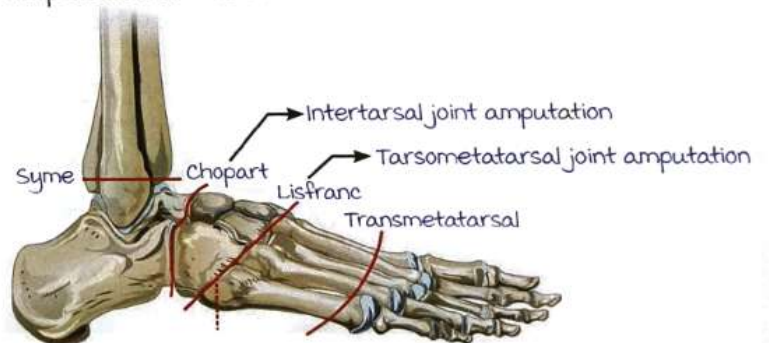
Lower limb :



Gritti-Stokes amputation :



Amputation of the foot :



Complications of amputations :

1. Amputation neuroma :

- Constant pain at amputation site.
- Cause : When nerves are not stretched & cut under tension → they do not retract into soft tissues → form neuroma.
- Rx : **Excision (best)**, Transcutaneous electrical nerve stimulation (TENS), Interferential therapy.

2. Phantom limb (seen in 50-80% cases) :

- Patient **feels** amputated limb is **attached** to the body.
- Sensation disappears in a distal to proximal fashion.

3. Infection.

4. Bleeding.

5. Flap necrosis : D/t tight suturing.

Prosthesis & Orthosis

00:30:59

Prosthesis :

- Device that replaces normal anatomy.
- E.g : Artificial limb after amputation.

Orthosis :

- Device that supports normal anatomy.
- E.g : Cock up splint for wrist drop.

PARTS

Socket : where amputation stump attaches

Connecting rod

Knee : Point where connecting rod meets terminal device

Terminal device

Types of terminal device :**1. Jaipur foot :**

more suitable for Indian population

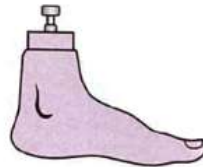


Flexible heel (allows movement)

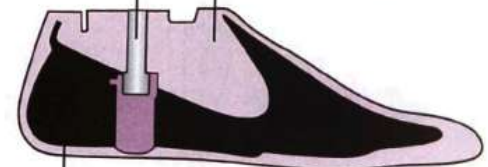
Wood

vulcanized
rubber
coating

Rubber core

**2. Solid ankle cushion heel (SACH) foot :**

Solid ankle block Wooden keel (Rigid)



Compressible heel (Cushioned)

Prosthesis	SACH	Jaipur foot
Appearance	Does not look normal	Looks normal
Walking barefoot and on uneven surfaces	Not possible	Possible
mobility	Restricted	Allowed
Dorsiflexion	Absent	Present
Inversion/Eversion	Absent	Present
Squatting	Not possible	Possible
Cost	High	Low

Note :**Amputations in children :**

- Save as much bone as possible.
- Disarticulation is preferred over amputation : It does not disturb bone growth by protecting the growth plate.
- Multiple revisions of stump is required.
- Children adapt well to prosthesis.

Reimplantation of Limb

00:34:16

----- Active space -----

Prerequisite : Clean wound with low MESS score.

Transportation of appendage :

- Clean in sterile NS or RL.
- Wrap in sterile gauze piece soaked in NS or RL.
- Keep in a plastic bag and seal it.
- Then keep it in an ice box (↑ Ischaemia time to 12-24 hrs).
- Then keep it in a box.

Order of reconstruction :

Mnemonic : BE FAN of virender sehwaq.

1. Bone.
2. Extensor tendons.
3. Flexor tendons.
4. Artery.
5. Nerve.
6. Vein.
7. Skin.

Advanced trauma life support (ATLS)

00:35:30

Simultaneous diagnostic and therapeutic activities intended to identify and treat life and limb threatening injuries beginning with the most immediate.

Indications :

Non-cardiac arrest patients who have had polytrauma.

Objective : To prevent death in the golden hour of trauma.

Components :

1. Airway :
Chin lift/jaw thrust with restriction of cervical spine motion.
2. Breathing.
3. Circulation (stop the bleeding).
4. Disability or neurological status.
5. Exposure (undress) and environment (temperature control).

Pelvic fracture management :

Can cause 1.5 - 2L of blood loss → hemorrhage & death.

1. Pelvis binder/hands/bedsheets
Tamponade the blood loss.
2. IV fluids : RL > NS.
3. External fixation of pelvis.



Stabilises the cervical spine



Philadelphia collar

----- Active space -----

Damage control orthopaedics

00:41:37

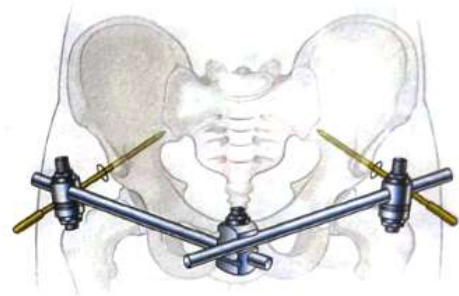
Limited surgical intervention to control hemorrhage & contamination.

Indications : unstable patients (Do not take them up for Sx → Sepsis and death)

Eg : Fat embolism syndrome, chest injury, coagulopathy, hypothermia, severe acidosis.

Steps :

1. Early temporary fixation using external fixator.
2. Patient is optimized (in ICU).
3. Definitive treatment.



External fixation of pelvis

Crush Syndrome

00:43:36

Pathogenesis :

Compression of limb for prolonged period



Hypoperfusion, myonecrosis



release of compression



Toxic metabolites get released



Reperfusion injury (Systemic manifestations of toxic metabolites)

myonecrosis → acute tubular necrosis, renal failure
 → metabolic acidosis, hyperkalemia, hypocalcemia

management :

1. Fluids :

Encourage high urine flow.

2. Alkalinization of urine with sodium bicarbonate :

Prevents myoglobin precipitation in renal tubules.

METABOLIC DISORDERS OF BONE

----- Active space -----

Diseases affecting development of bone.

Causes bone weakness $\rightarrow$ microscopic fractures (pain) $\rightarrow$ further weakens bone $\rightarrow$ Pathological fracture.

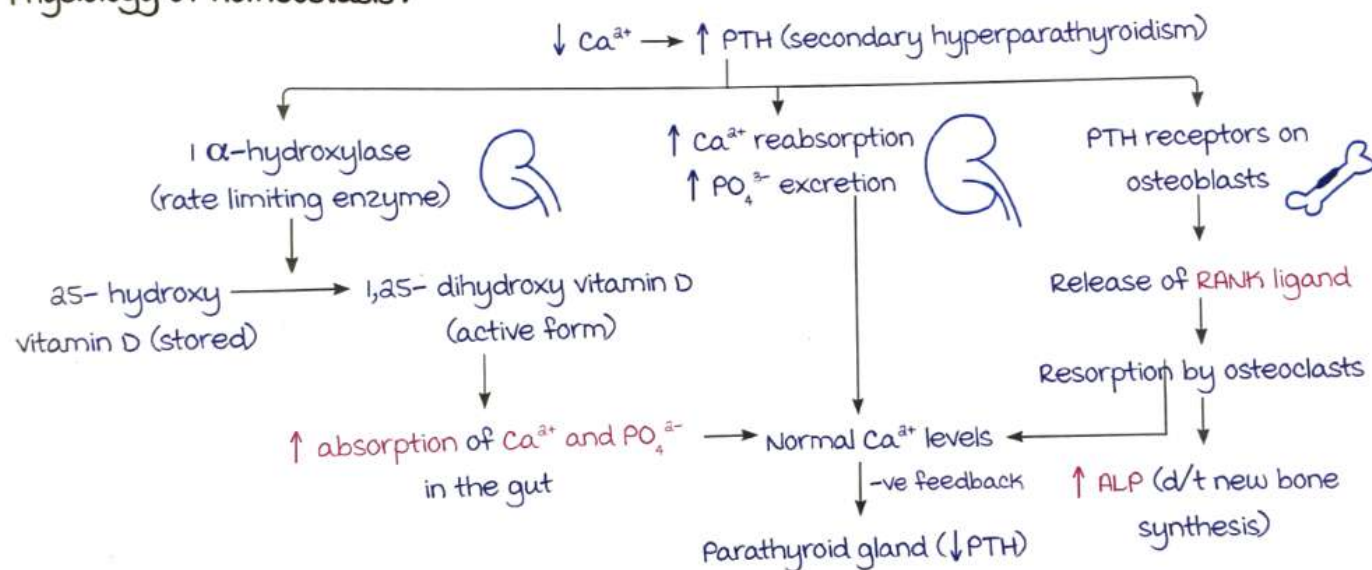
Calcium Homeostasis

00:00:30

maintained by: $\begin{cases} \text{Parathyroid hormone (PTH).} \\ \text{Vitamin D} \end{cases}$

HYPOCALCEMIA

Physiology of homeostasis:



Causes: vitamin D deficiency (rickets/osteomalacia).

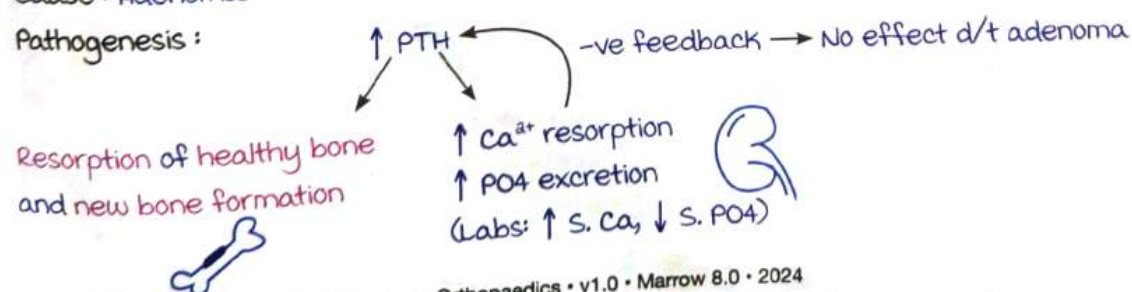
Lab parameters: $\uparrow$ /Normal Ca^{2+} , PTH, PO_4^{3-} , ALP $\uparrow$.

Note: Denosumab $\xrightarrow{(-)}$ RANK ligand $\rightarrow$ inhibits osteoclast activity.

Primary hyperparathyroidism:

Cause: Adenoma.

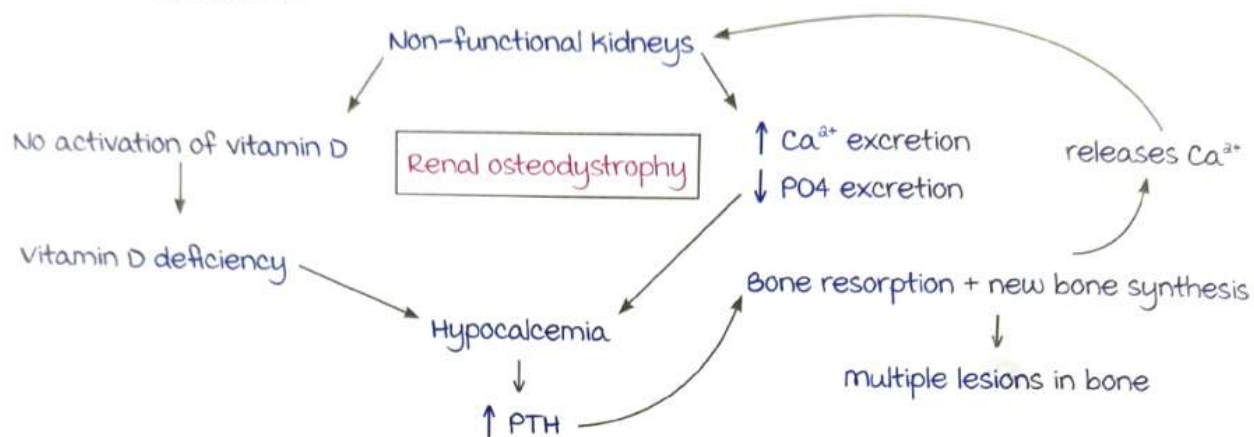
Pathogenesis:



----- Active space -----

Renal failure :

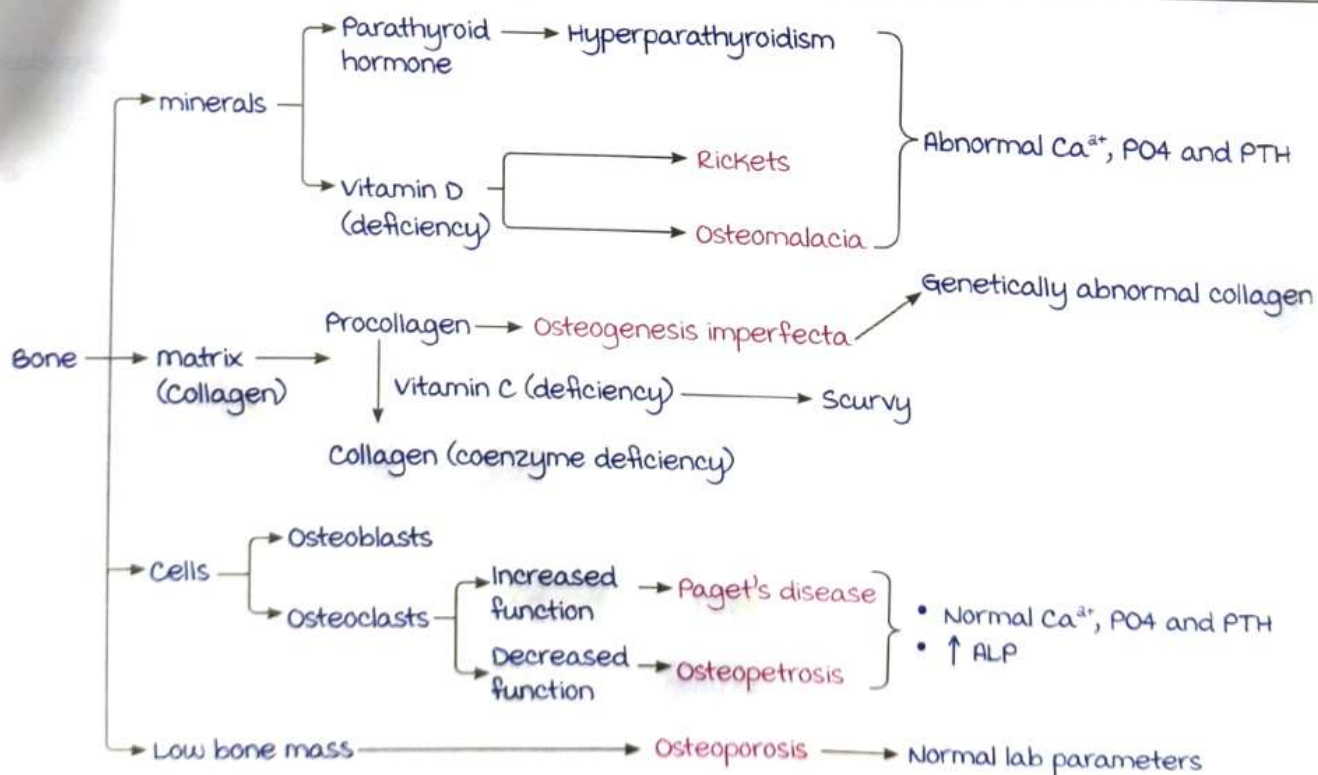
mechanism :

**Laboratory picture in hyperparathyroidism :**

Hyperparathyroidism		Ca ²⁺	PO ₄	PTH	ALP
1. Primary		↑	↓	↑	↑
2. Secondary	d/t vitamin D deficiency	↓/Normal	↓	↑	↑
	d/t renal failure	↓	↑	↑	↑

Overview of metabolic bone disorders

00:13:22



Rickets

00:18:17

----- Active space -----

CAUSES

- vitamin D deficiency : m/c cause
 - Nutritional (m/c cause).
 - malabsorption.
 - Lack of sunlight exposure.
 - Liver and kidney disease.
 - Drugs.
- Ca^{2+} deficiency.
- PO_4 deficiency.

LABS

↓ / normal Ca^{2+} , ↓ PO_4 -, ↑ PTH, ↑ ALP.

CLINICAL FEATURES

1. Skull :

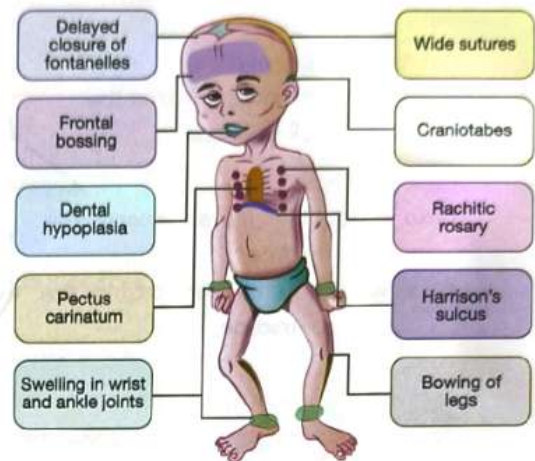
- **Craniotabes**/softening of skull/ping pong skull : Earliest change.
- Frontal bossing.
- Delayed closure of fontanelle.

2. Chest :

- **Rachitic rosary** : Costochondral junction swelling (blunt and non-tender).
- **Pigeon chest**/pectus carinatum : Prominent sternum.
- **Harrison sulcus** : Due to bending of softened ribs by diaphragm contraction.

3. Long bone deformities :

- Only seen once child starts weight bearing.
- Not seen in infants.



B/L genu varum
(m/c presentation)



B/L genu valgus



windswept deformity

Valgus on one side
Varus on one side

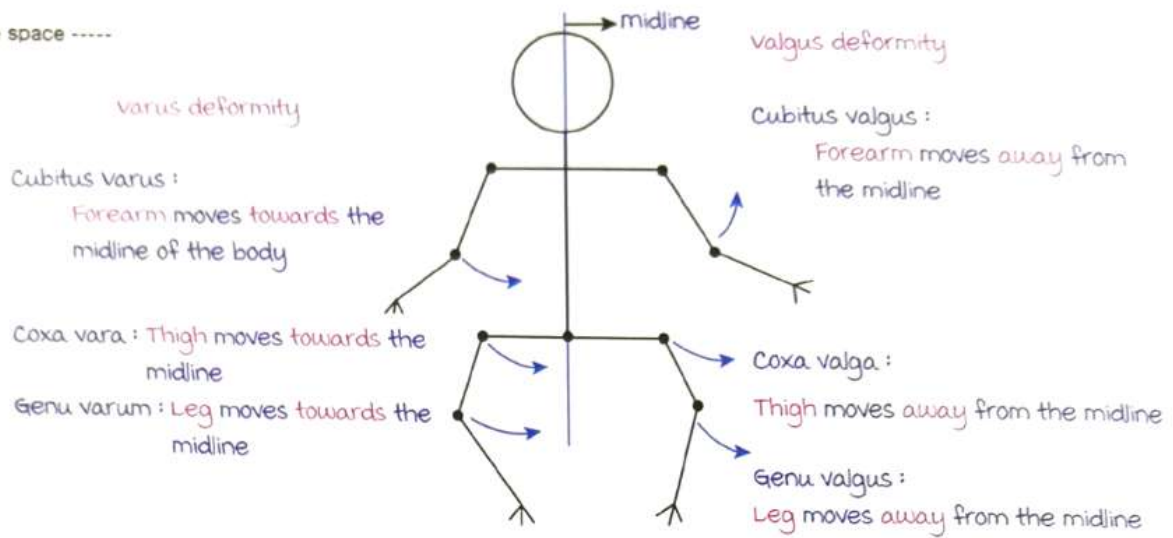


B/L genu varum

Note :

Scorbutic rosary (d/t vitamin C deficiency) : Painful as they are due to bleeding.

----- Active space -----



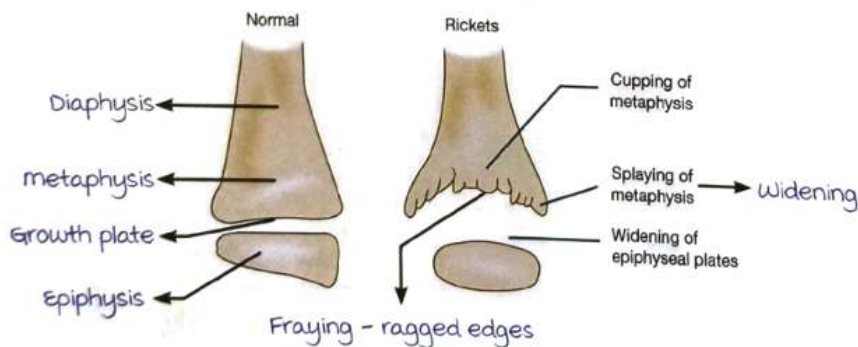
	B/L genu varum (Bow legs)	B/L genu valgus (Knock knees)	Windswept deformity
m/c cause in children	Rickets > Idiopathic	Idiopathic > rickets	Rickets
m/c cause in adults	OA > RA	RA > OA	RA

Genu varum seen in a child more than 2 yrs of age → Rickets.

Deformities at the hip :

- Coxa vara ($<120^\circ$)
- Normal ($120-135^\circ$)
- Coxa valga ($>135^\circ$)

RADIOLOGICAL FEATURES



Healing rickets :

- Best method to assess healing of rickets is X-Ray and serum ALP.
- Ca^{2+} normalises
- PTH ↓
- PO_4^{3-} ↑
- ALP ↓

White line of Frankel
(mineralization of growth plate)



Healing rickets

TREATMENT

- Young children : Deformities are reversible upon starting treatment.
- Older child : Osteotomy for residual deformity after bone activity is reduced.

----- Active space -----

Osteomalacia

00:31:44

FEATURES

- Female > males.
- Young people.

Symptoms :

- Poly arthralgia.
- Bone pains.
- Proximal myopathy.

INVESTIGATIONS

Radiological :

X-Ray :

1. Pseudofractures :

- AKA *milkman's fracture/looser zone*.
- m/c site : Neck of femur, clavicle
Ribs, scapulae, pubic rami.



Transverse band of rarefaction

2. Protrusio acetabuli :

Head of femur pushes into the acetabulum.



3. Champagne bottle deformity :

Also seen in achondroplasia.



4. Triradiate pelvis.

5. Trefoil pelvis.

Biopsy : Diagnostic.

Note :

Other conditions where *looser's zone* can be seen :

- Hypothyroidism.
- Paget's disease.
- Fibrous dysplasia.
- Osteogenesis imperfecta.
- X linked hypophosphatemic rickets.

----- Active space -----

TREATMENT**Vitamin D :**

1. **Stoss regimen** : 3 lakh - 6 lakh IU deep IM/oral (stat or over 1-5 days).
2. Daily 2k - 5k IU for 4-6 weeks.
3. Weekly 50k - 60k IU for 8-12 weeks.

Note :• **Tumor induced osteomalacia :**

- Fibrosarcoma.
- Osteoblastoma.
- Osteosarcoma.
- Non-ossifying fibroma.

• **X-Ray scanogram :**

- Full length X-Ray of weight bearing lower limb.
- measures and quantifies deformity.

Scurvy

00:39:38

Collagen maturation and osteoid formation defect d/t vitamin C deficiency.

Features of scurvy**Bone (Radiological features)**

- Diaphysis
 - Ground glass appearance
 - Pencil thin cortex
- metaphysis
 - Whiteline of Frankel
 - Scorbutic zone
 - Pelkan's spur
- Epiphysis
 - Wimberger Ring Sign

D/D :

- Healing rickets
- Congenital syphilis
- Plumbism
- Leukemia

**Blood vessel (Symptoms)****Bleeding Gums****Costo Chondral Junction :**

Scorbutic Rosary

Bleeding gums**Bone :****Subperiosteal hemorrhage**

B/L knee pain

**Findings in Scurvy**

Treatment : Vitamin C.

----- Active space -----

Note :

Barton disease :

- Vitamin C and D deficiency.
- Features of both are seen.

Hyperparathyroidism

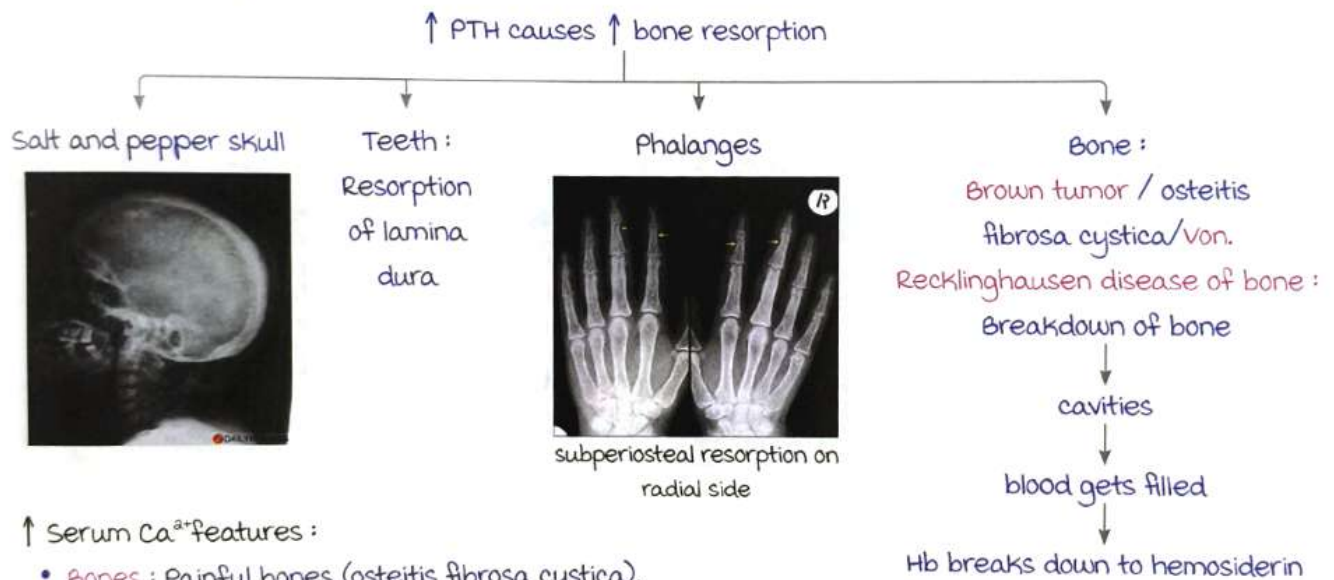
00:42:41

1° hyperparathyroidism : $\uparrow$ PTH irrespective of Ca^{2+} levels.

- m/c cause : adenoma.
- Rx : surgical excision of adenoma.

2° hyperparathyroidism :

- Hypocalcemia.
- Vit D deficiency.
- Renal failure.
- Liver failure.



$\uparrow$ Serum Ca^{2+} Features :

- **Bones** : Painful bones (osteitis fibrosa cystica).
- **Stones** : kidney stones.
- **Groans** : Abdominal symptoms.
- **Psychiatric overtones** :
 - Nervous system : Lethargy, fatigue, memory loss, depression, psychosis.

Note :

Renal osteodystrophy X-Ray :



Rugger Jersey spine

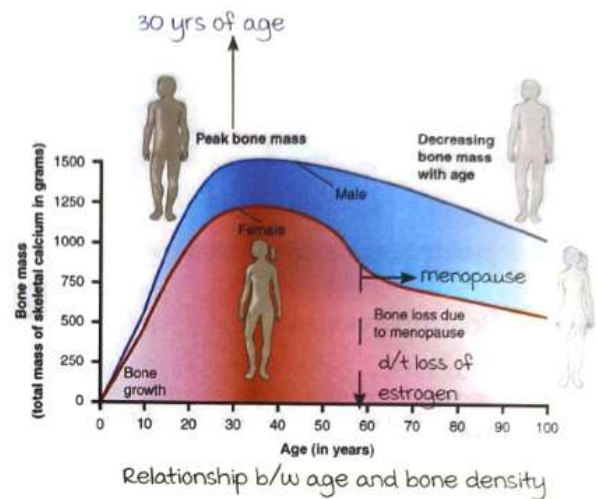
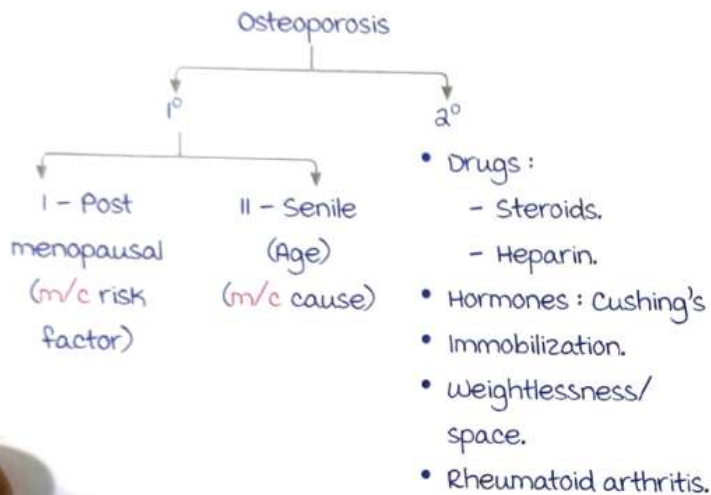
→ Alternating zones of $\uparrow$ and $\downarrow$ bone density

----- Active space -----

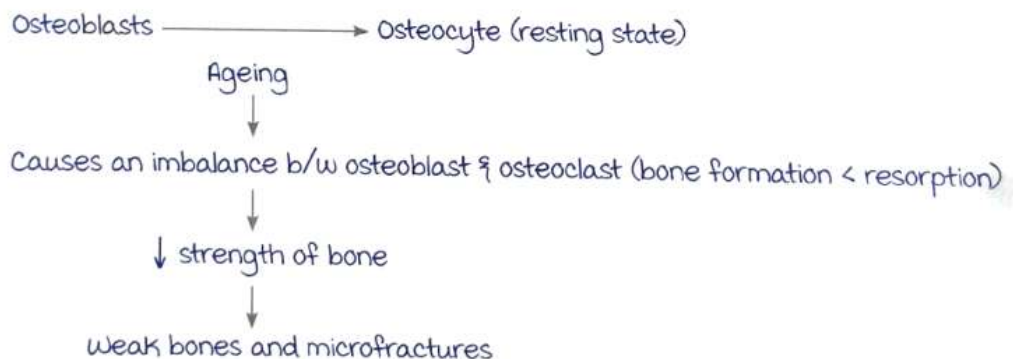
Osteoporosis

00:47:09

- Porous defect of bone.
- Normal quality, ↓ quantity.

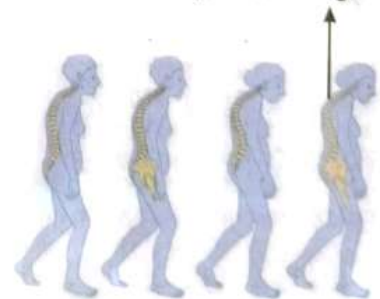
ETIOLOGY

- Estrogen: Has a protective effect against osteoporosis.

PATHOGENESIS**CLINICAL FEATURES**

- Back pain: **Earliest symptom.**
Pain before fracture (constant weight bearing).
- Fragility fracture:
 - m/c complication.
 - Sites: Spine, Neck of femur/ hip, colle's fracture.
- Deformity: **Kyphosis.**

Forward bending of spine: kyphosis



Deformity d/t spine fracture

Lab findings :

Normal.

----- Active space -----

Radiological features :

x-ray changes are only seen after 30 % bone loss.



Spine : Lateral view



Neck of femur fracture



Cod fish vertebrae/Fish mouth vertebrae/Biconcave vertebrae



Colle's fracture

Note :

D/D for cod fish vertebrae :

- Osteogenesis imperfecta
- Osteomalacia

SCREENING

Prerequisites/ criteria to screen :

- Women age $\geq$ 65 yrs.
- Men of age $\geq$ 70 yrs.
- Fractures seen after 50 yrs of age.

----- Active space -----

DEXA scan (Dual energy X-ray absorptiometry) :

- 100
- Assesses spine (preferred), hip & calcaneum.

T score :

- Density is compared with a population having peak bone density.
- Young reference male/ female, 30 yrs of age.
- WHO classification of osteoporosis (based on T-score).
 - 0 to -1 : Normal.
 - -1 to -2.5 : Osteopenia.
 - < -2.5 : Osteoporosis.
 - < -2.5 with fracture : Severe osteoporosis.

TREATMENT**Drugs ↓ bone resorption :**

Inhibit osteoclastic activity.

1. Bisphosphonates : First line drugs.
 - Risedronate, Ibandronate, Zoledronate.
 - Adverse effects :

Prolonged use of bisphosphonates (5 - 7 yrs)

↓ osteoclastic activity
(lower than osteoblastic activity)

Atypical fractures around hip

- Drug holiday : Temporarily stop using bisphosphonates & switch to other drugs (Teriparatide).
2. Denosumab : monoclonal Ab against RANK ligand.
 3. SERM : Raloxifene.
 4. HRT : For post menopausal symptoms.

Drugs that ↑ bone formation :

↑ activity of osteoblast (anabolic).

1. Teriparatide : Synthetic PTH (small doses increase only osteoblasts).
 - First line in severe osteoporosis.
 - used for shorter duration.
 - Adverse effect : Cancer in rat models.
2. Abaloparatide : Synthetic PTH.
3. Romosumab : Anti sclerostin Ab (Sclerostin → Osteoblasts)



Atypical fracture

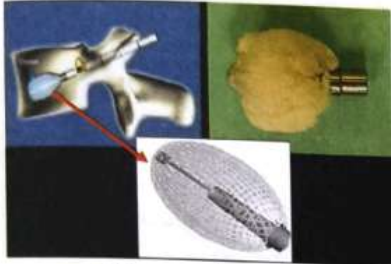
Drugs that ↓ bone resorption & ↑ bone formation :

- Strontium Ranelate : ↑ complications.

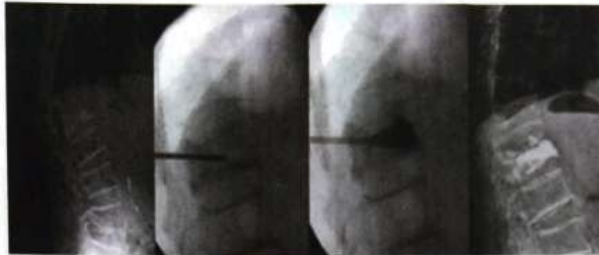
----- Active space -----

Treatment of deformities :

- vertebroplasty : Inject PMMA bone cement.
- Balloon kyphoplasty : Restores height of vertebra → Inject bone cement.



Balloon Kyphoplasty



Vertebroplasty

Paget's Disease/ Osteitis Deformans

01:07:43

- male > female.
- Incidence : 5th decade.

Etiology :

- Idiopathic (m/c).
- SQSTM1 gene mutation.
- Paramyxovirus.
- Pelvis > tibia.

Pathogenesis :

↑ osteoclast activity → ↑ bone resorption → ↑ bone formation by osteoblasts
(↑ ALP)

Stages :

- Lytic phase.
- Mixed phase.
- Blastic phase.

Clinical features :

1. Pain (m/c).
2. Bone is warm to touch & thickened.
3. Weak bone (Banana fracture).
4. Cranial foramen stenosis : CN 2, 5, 7, 8 (hearing disturbances).
5. Otosclerosis.
6. Thickening of skull : Frequent changes in hat size.

Investigations :

- Biopsy (IOC) : mosaic pattern.
- ↑ ALP.



Banana fracture

----- Active space -----

Radiological features :

1. Lytic phase :

- Blade of grass/flame shaped appearance.
- Osteoporosis circumscripta : Circumscribed lesion in the skull.



Blade of grass deformity

2. mixed phase :



Picture frame vertebrae



Ivory vertebrae



Cotton wool skull

3. Blastic phase :

Thickening of skull



Tam O shanter skull

Treatment :

1. Zoledronate :

- Long acting bisphosphonate.
- Suppress osteoclastic activity.

2. Calcitonin : To manage pain.

Complications :

- High output cardiac failure (causes death).
- Transformation to osteosarcoma (d/t ↑ bone turnover).

Osteogenesis Imperfecta/ Brittle Bone Disease

01:14:27

- Genetically abnormal collagen.
- COLA I gene defect.

Types :

1. AD/ Type I/ Classical :

Weak bone → Recurrent pathological fractures.

2. AR/ Type II : (Still born/lethal form)

multiple fractures in utero (seen in antenatal scan).

Investigations :

Normal laboratory values.

Clinical features :

1. Deformities : D/t multiple fractures in the same bone.



2. Blue sclera :



----- Active space -----

Choroid is visible d/t thin sclera

3. Easy bruising.

Treatment :

1. Braces :

Prevention of fractures.

2. Realignment osteotomies :

Seekh Kabab operation.



Laboratory Values in Metabolic Bone Disorders

01:16:48

	Calcium	PTH	PO4	ALP
2° hyperparathyroidism/ Rickets/ Osteomalacia	↓ / Normal	↑	↓	↑
Renal osteodystrophy	↓↓	↑	↑↑↑	↑
Osteoporosis	Normal			
1° hyperparathyroidism	↑↑↑	↑	↓	↑
Paget's disease	Normal			↑↑↑

----- Active space -----

UPPER LIMB TRAUMA : CLAVICLE AND SHOULDER

mechanism of Injury :

Fall on an outstretched hand.



X-ray shoulder : AP view

Clavicle

Acromion

Proximal humerus

Lateral border of scapula

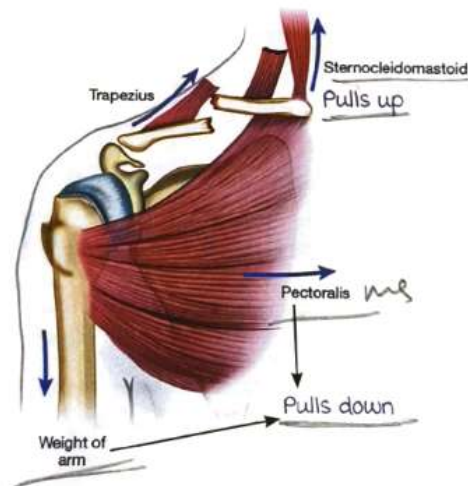
Fracture of Clavicle

00:04:01

Clavicle most commonly fractured at medial 1/3 - tubular. lat. 2/3 - flat.

- m/c fracture
 - At birth
 - In newborn
 - Overall : D/t anatomy
 - medial 1/3rd : Tubular.
 - Lateral 2/3rd : Flat.
- m/c site of fracture : Junction of medial 2/3rd (3/5th) and lateral 1/3rd (2/5th).

DISPLACING FORCES IN FRACTURE CLAVICLE



Lateral fragment : Pushed down

medial fragment : Pulled up



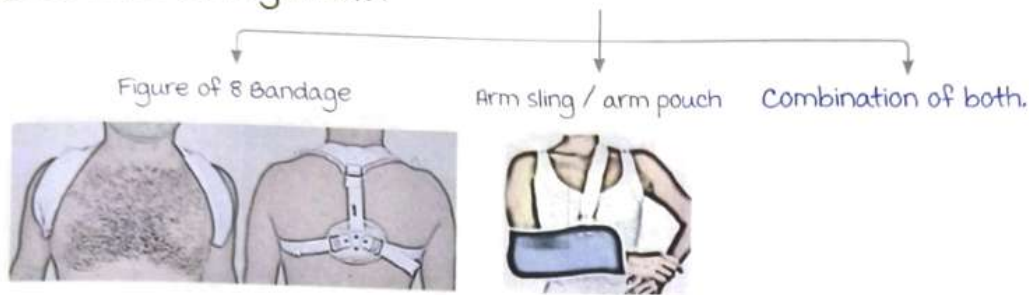
Fracture of Clavicle

COMPLICATIONS :

- malunion (m/c) : D/t displacement.
- Neurovascular injury to
 - Brachial plexus.
 - Subclavian vessels.
- Non-union (Very rare).

MANAGEMENT**Conservative management :**

----- Active space -----

**velpeau bandage :**

- Indications →
- Fracture of clavicle.
 - Acromion injuries.
 - Shoulder dislocations.

**Surgery :****Indications :**

1. Open clavicle fracture.
2. massive displacement (Skin tenting +).
3. Acromioclavicular joint involved.
4. Neurovascular injury.
5. Floating shoulder injury (Clavicle fracture + glenoid fracture).

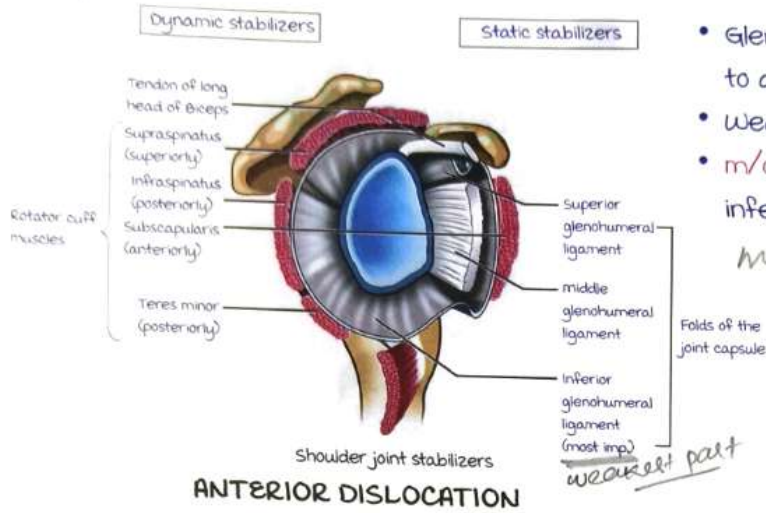
Procedure : Open Reduction + Internal Fixation.**Shoulder injuries**

00:09:36

ANATOMY OF SHOULDER JOINT

- Type of joint : Ball (Head of humerus) & socket (Glenoid) joint.
- Size of head : 4x size of socket
 - High mobility → most commonly dislocated.
 - Low stability.

----- Active space -----



- Glenoid labrum: Provides some concavity to accommodate head of humerus.
- Weakest point of shoulder joint: Inferior.
- m/c dislocation: Anterior > posterior > inferior.

Most mobile & m/c dislocation is

Subtypes:



Pre glenoid



Subcoracoid (m/c)



Subclavicular



Intrathoracic

Presentation:

Attitude of limb (Abnormal position of limb after dislocation).

- Loss of shoulder contour → Flattened.
- Abduction & external rotated → Loss of adduction & internal rotation.

Arm by side of the body

Mechanism of injury → H/o trauma.

→ Fall on outstretched hand.

Acute dislocation: Emergency → Requires immediate reduction.

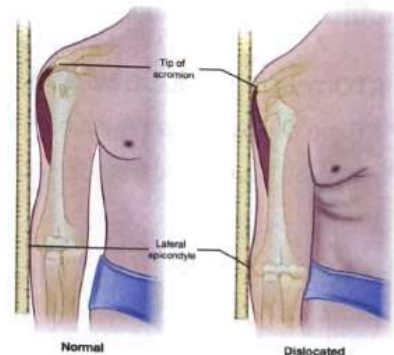
On examination:

1. Hamilton Ruler Test

Ruler placed against lateral epicondyle of humerus

↓

Ruler touches acromion d/t displaced head of humerus



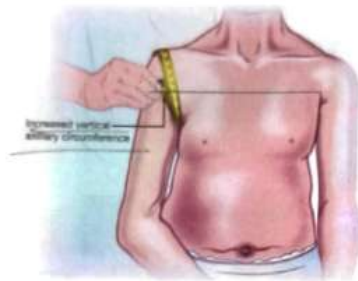
Loss of shoulder contour in anterior dislocation

2. Dugas test



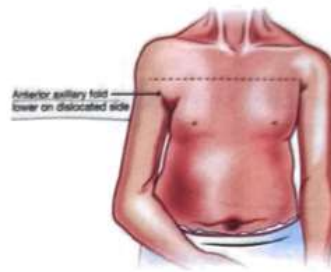
unable to touch
opposite shoulder

3. Callaway test



Increased vertical
axillary circumference

4. Bryant's test



Anterior axillary fold
lower on dislocated side

----- Active space -----

5. Axillary nerve injury :

Loss of → motor function of → **Deltoid**
→ **Teres minor.**

→ Sensory function : **Regimental badge sign** (Paresthesia/tingling/
numbness/tenderness over lateral aspect of
upper arm).

X-ray :

- AP view.
- Lateral view.



X-ray : Anterior dislocation of shoulder joint

Techniques of Reduction :

1. modified Kocher's Technique (**TEAM**) : m/c to be used.



1. Traction



2. External rotation



3. Adduction

4. medial / Internal
rotation

----- Active space -----

Note :

Attempt closed reduction under analgesia

Fails d/t muscle spasm

Attempt closed reduction under anesthesia (Causes muscle relaxation)

Fails

Open Reduction

2. Stimson's technique :



Patient prone

Heavy object tied to dislocated limb

Hang limb at the edge of table/bed

Gradual traction overcomes muscle spasm

Reduction achieved

After reduction

Place arm in arm pouch/ sling/ velpeau bandage

Promote healing of soft-tissue injuries

↓ risk of recurrent dislocations

3. Hippocratic technique :

• Countertraction :
with foot

• Traction : With arm

Stabilise pt by
arm, arm pouch,
sling, velpeau bandage

Complications of recurrent dislocations :

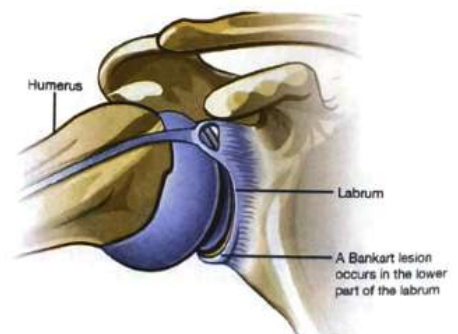
1. Bankart lesion (m/c).

Glenoid labrum tears antero-inferiorly

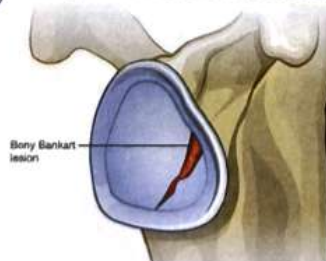
Laxity of inferior glenohumeral ligament

↓ stability

Recurrent dislocations



• Bony bankart : Avulsion fracture of antero-inferior glenoid

Due to tear of
inferior glenohumeral
lig.

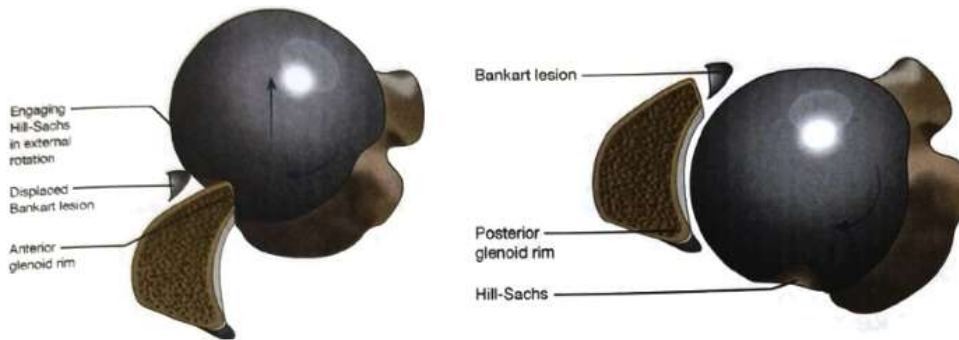
Bony bankart

2. Hill-Sachs lesion

Recurrent dislocation → Head of humerus repeatedly hits anterior glenoid rim

----- Active space -----

Indentation on posterolateral surface



mechanism of Hill-Sachs lesion



X-ray : Hill-Sachs lesion

Note :

- m/c complication of shoulder dislocation : Recurrent dislocation.
- m/c → Early complication : Axillary nerve injury
- Late complication : Recurrent shoulder dislocation
- m/c nerve injured in shoulder dislocation : Axillary nerve.

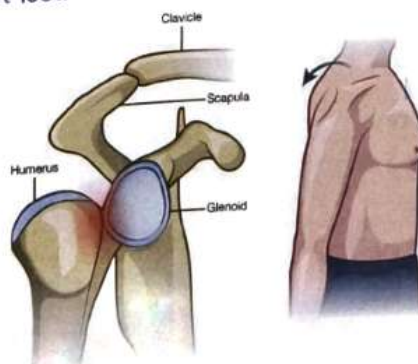
POSTERIOR DISLOCATION

mechanism of injury :

1. High velocity muscular violence (Specific)
 - In seizures.
 - High voltage electric shock.
 - In Electroconvulsive therapy (ECT).
2. Fall on outstretched hand
3. Trauma

Attitude of limb :

- Adducted and internally rotated
- Loss of abduction & external rotation
- Shoulder contour : Not lost.
- Pain (+).



----- Active space -----

X-ray :

Electric light bulb sign.



Normal X-ray



X-ray in posterior dislocation

Complications :

- Lesions
 - Reverse Bankart : Postero-inferior tear of glenoid labrum.
 - Reverse Hill-Sachs : Antero-medial indentation of head of humerus.
- Nerve injury : Axillary nerve (m/c).

INFERIOR DISLOCATION :AKA *Luxatio Erecta*.

mechanism : Hyperabduction injury.

Note :

Tests for shoulder instability.

Instability	Test
Anterior	Fulcrum
	Crank
	Apprehension
Posterior	Jerk
Inferior	Sulcus

Treatment of unstable shoulder : Surgery.

- Arthroscopic
 - Bankart's repair
 - Hill Sach's repair
 - Bristow Latarjet operation
 - Putti-platt operation
- } Obsolete

UPPER LIMB TRAUMA : ARM AND ELBOW

----- Active space -----

Proximal Humerus Fracture

00:00:08

- Risk factor : Postmenopausal osteoporotic females (rate of osteoporosis : In proximal humerus > other parts).
- m/c site : **Surgical neck of humerus** (passes through lesser & greater tuberosities).
- m/c injured nerve : **Axillary nerve**.

78

NEER'S CLASSIFICATION



Fracture : Surgical neck of humerus

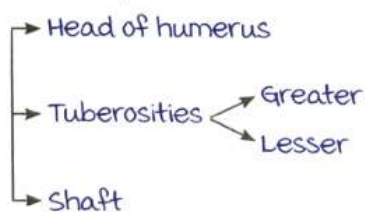


NEER's type 2 proximal humerus fracture

Depending on the parts (fragments) displaced from normal anatomy :

Type	No. of fragments
I	0 (undisplaced)
II	2
III	3
IV	4

Parts of proximal humerus :



No. of parts $\propto$ risk of complications (E.g : Avascular necrosis).

TREATMENT

Definitive :

Open Reduction (OR) + Internal fixation (IF) with plates & screws.



Fracture



OR+IF with plates & screws

----- Active space -----

Shaft of Humerus Fracture

00:05:14

- Fracture between surgical neck of humerus & supracondylar area.
- Associated injured nerve : Radial nerve (runs in radial groove on the shaft of humerus).

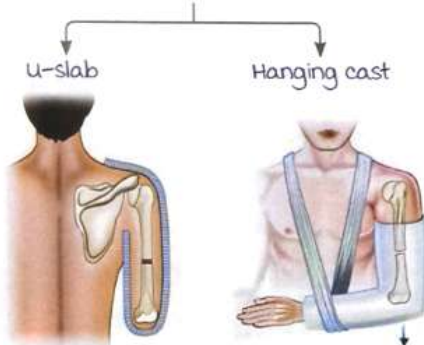
Treatment :

Closed reduction (CR)

+ POP application

Open Reduction (OR)

+ IF with plates & screws } Best



Note :

Hanging cast : Keeps fracture reduced with gravity → Requires patient to be in upright position to work.

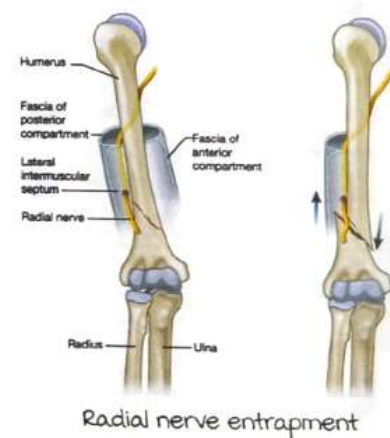
Holstein-Lewis Fracture :

Site : Junction of upper 2/3rd & lower 1/3rd
(radial nerve perforates lateral intermuscular septum)

↓
Causes entrapment of tethered radial nerve

↓
Wrist drop + sensory loss

↓
Rx : Cock-up splint



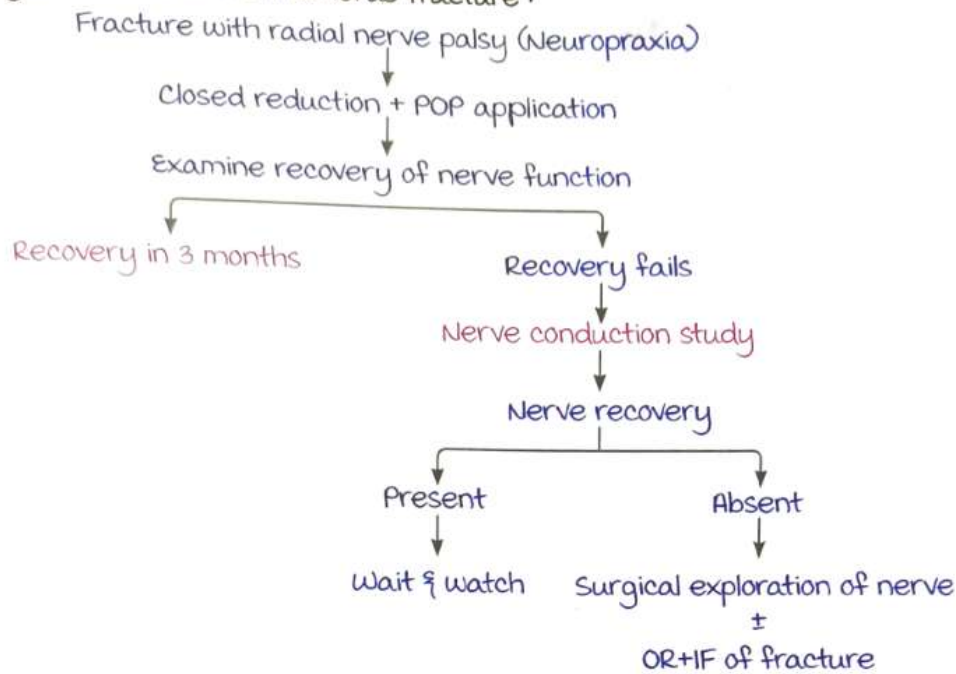
Holstein-Lewis Fracture



Cock-up splint

----- Active space -----

management of shaft of humerus fracture :

**Fractures around elbow**

00:18:23

Types :

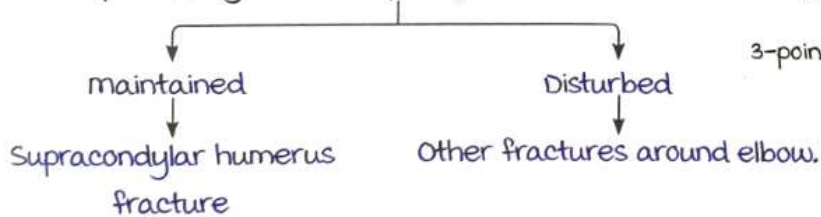
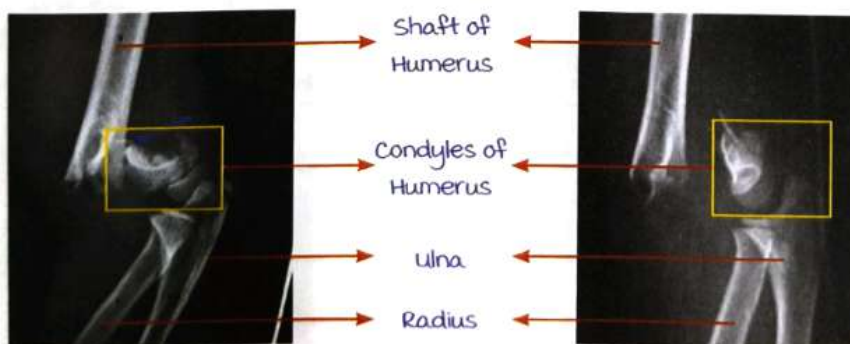
- Supracondylar humerus (m/c).
- Lateral condyle fracture (and m/c).
- medial condyle fracture.
- Transcondylar fracture.
- Intercondylar fracture.
- Olecranon fracture.

Note :

Elbow joint : hinge jt
ulnohumeral joint.

3-point Bony relationship in elbow

Three point bony relationship in elbow :

**SUPRACONDYLAR HUMERUS FRACTURES**

----- Active space ----- Salter Harris type 2 (extraarticular) fracture :

- m/c fracture around : elbow in children : D/t anatomy

Soft, thin, narrow bones in children Flattened supracondylar area Thin d/t olecranon & coronoid fossa

- m/c mode of injury : Fall on an outstretched hand
 - m/c mechanism of injury : Hyperextension
- } → Posterior displacement.

Types based on displacement of distal fragment :

Extension type
(posterior displacement)

Flexion type
(anterior displacement)

mechanism

Fall on extended elbow
+
Triceps pull

Displacements in extension type

1. Proximal migration
 2. medial tilt and medial shift
 3. Internal rotation
 4. Posterior (dorsal) tilt & shift
- } m/c

Gartland Classification :

	Type I	Type II	Type III
Appearance	Undisplaced (fracture line may not be visible)  fat pad sign	• Anterior cortex : breached • Posterior cortex : intact (incomplete fracture). 	Displaced complete fracture  Fish tail sign
Definitive treatment	POP applied in 90° flexion of elbow	Reduction ↓ Adequate Inadequate ↓ ↓ POP K wires applied inserted	Open/closed reduction + IF with K-wires

• **Fat pad/Sail sign:**

Hematoma around fracture pushes away fat
 ↓
 Sail-shaped lucency around supracondylar area

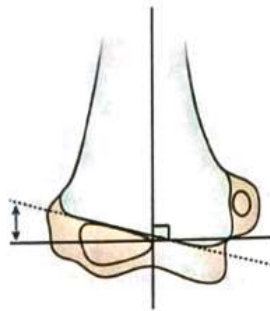


Fish-tail deformity

----- Active space -----



Dunlop traction
 (For temporary stabilisation of fracture)



Baumann's angle
 (to determine adequate fracture reduction)



ORIF of supracondylar fracture

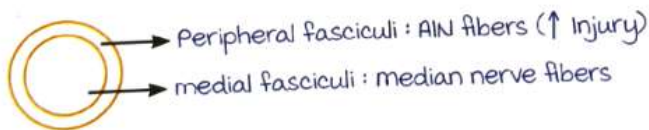
Complications:

Early (neurovascular)

- vessel injury : m/c → **Brachial artery**
- Compartment syndrome
- Volkmann Ischemic contracture
- Nerve injury : m/c → **Anterior interosseous** > median > Radial

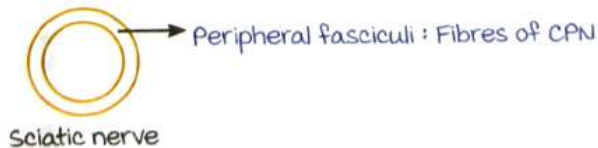
Late

- malunion (m/c) → cubitus varus / gunstock deformity (↓ carrying angle)
- myositis ossificans

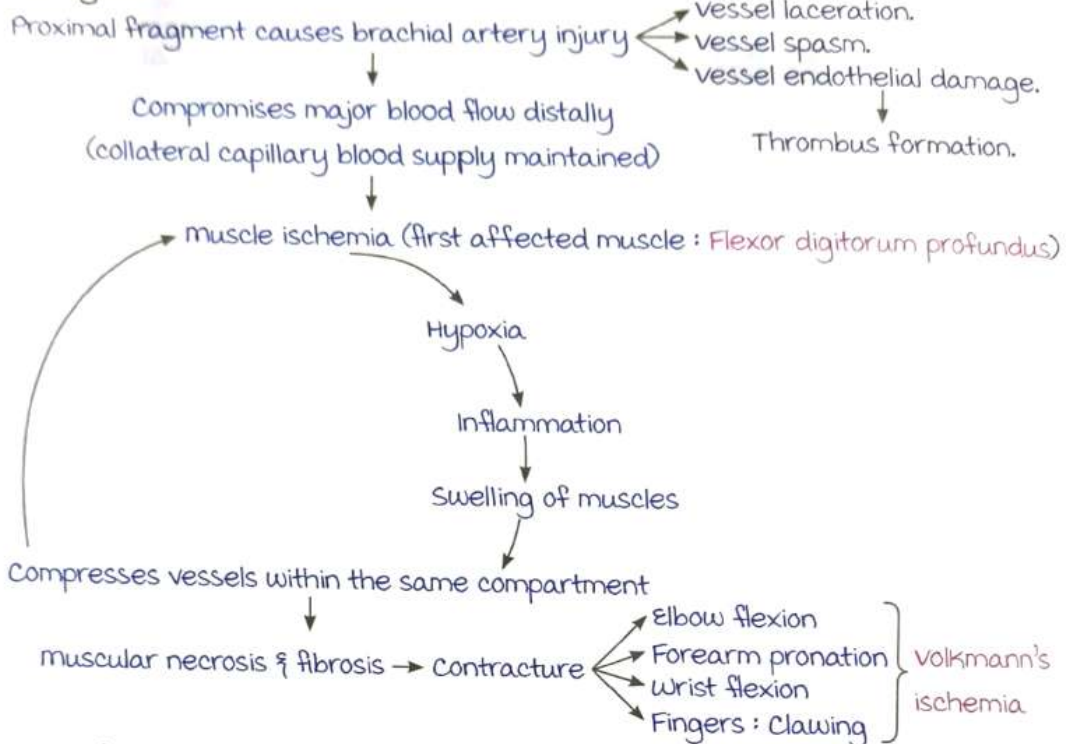


Cross-section of median nerve above elbow

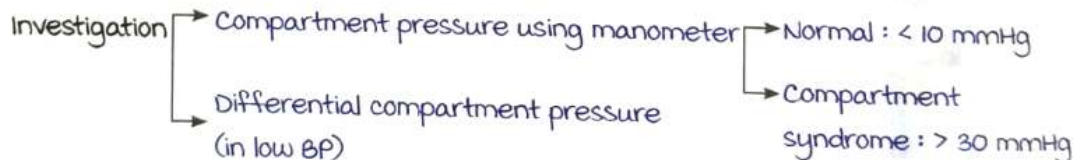
Note: Foot drop in posterior dislocation of hip : D/t common peroneal nerve (CPN) palsy.



----- Active space -----

Compartment Syndrome :**Pathogenesis :****Clinical features :**

1. Pain on passive stretch : most important sign.
2. Puffiness (Swelling).
3. Pallor.
4. Paraesthesia.
5. Paralysis (Late sign).
6. Pulselessness (very late sign).

**Treatment :**

Fasciotomy : Longitudinal incisions on superficial & deep fascia in all compartments

↓

Fall in compartmental pressure

↓

Improves vascularity → wound left open till pressure falls



Fasciotomy

After few days

- Skin graft
- vacuum/negative pressure therapy

Volkman's Ischemic Contracture :

Features :

- Deformity
 - Elbow flexion.
 - Forearm pronation.
 - Wrist flexion.
 - Fingers clawing.

- Sensory loss and motor paralysis in forearm & hand.

flexor group of muscles die

Treatment (Based on severity of contracture):

- mild : Turn buckle splint.
- moderate : max page (muscle sliding) operation.
- Severe : Bone shortening.



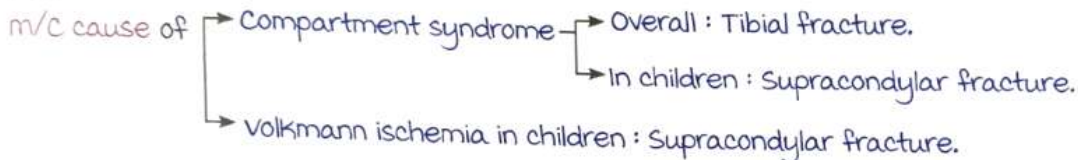
Volkman's Ischemic Contracture of upper limb

----- Active space -----



Turn buckle splint

Note :

**Pink Pulseless Hand :****Supracondylar humerus fracture**

Brachial artery compression / spasm

Collateral perfusion maintained

Radial pulse (-)

Limb appears pink

Rx : Fracture reduction (improves vascularity)

Malunion :

- m/c complication.
- D/t medial tilt + internal rotation of distal fragment → Gunstock / cubitus varus deformity.
- Extraarticular fracture → Elbow function : Normal.
- Rx : modified French osteotomy.

Note : Normal carrying angle

- : 5°.
- ♀ : 10-15°.

Myositis Ossificans :

- massage of fracture
 - Fracture hematoma dislodges
 - ↓ *brachialis*
 - Settles inside muscle
 - ↓
 - Bone forms inside muscle
 - ↓
 - ROM / restriction of movement



myositis Ossificans

----- Active space -----

- m/c muscle involved : **Brachialis**.
- m/c joint involved : **Elbow** > hip.
- Only cause for elbow ROM in supracondylar fracture.
- Rx → Active phase (pain, redness, swelling (+)) : NSAIDs, limb immobilisation.
- Latent phase : Active physiotherapy.
- ↓ Elbow function not restored
- Surgical excision of mass

Fat embolism syndrome :

Polytrauma/long bone fracture $\xrightarrow{24-48 \text{ hours}}$ Difficulty breathing, tachypnea, petechial rashes.

LATERAL CONDYLE HUMERUS FRACTURE

mechanism of injury : Fall on outstretched hand with slight varus force.

	Lateral condylar fracture	supracondylar fracture
Type of fracture	Intraarticular	Extraarticular
Salter-Harris classification Type	Type iv	Type i/ii
3-point bony relationship	Disturbed	maintained

Note :

Fracture of necessity (requires surgical management).

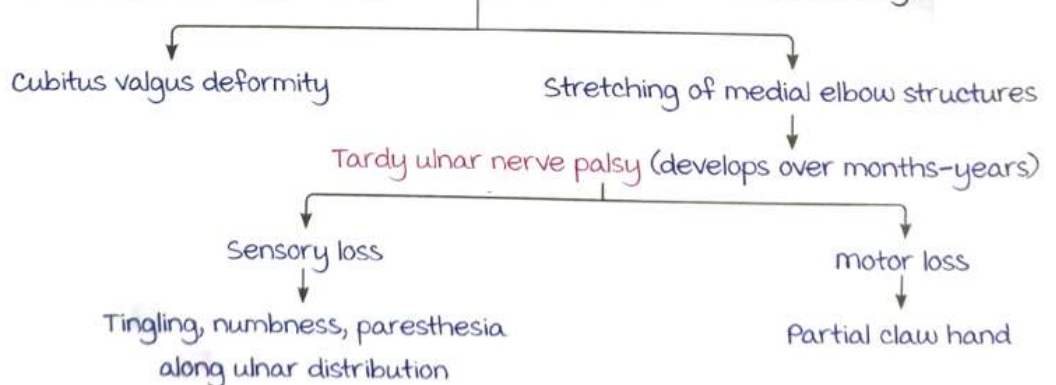
1. Intraarticular fractures.
2. Lateral condylar humerus.
3. Monteggia / Galeazzi.
4. Neck of femur.



Lateral condyle humerus fracture

Complications :

1. Elbow stiffness.
2. Non-union : D/t → Pull by common extensor group of muscles.
→ Compromised vascularity.
3. Damage to lateral growth plate → Growth of medial > lateral condyle.



Treatment : ORIF.

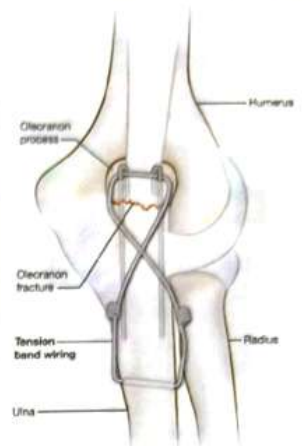
OLECRANON FRACTURE



X-ray appearance



Treatment



Rx : OR + IF with tension band wiring (TBW)
(Converts distractive forces → Compressive forces).

Elbow dislocation

01:03:01

- ulno-humeral dislocation.
- m/c dislocation in children.
- m/c type : Posterior.

Clinical features :

- 3-point bony relationship : Reversed.
- Attitude of limb : Flexion at elbow.
- Bow string sign : Posterior dislocation of elbow → Laxity of triceps muscle.



Elbow dislocation

Note :

Terrible (Hotchkiss) triad of elbow :

1. Posterior elbow dislocation.
2. Coronoid fracture.
3. Radial head fracture.

PULLED ELBOW / NURSEMAID'S ELBOW

Radial head pulled out of annular ligament.

----- Active space -----

mechanism :

In children <4-5 years : unossified smaller radial head in larger annular ligament

↓
Axial traction of extended & pronated elbow

↓
Radial head subluxates out

Clinical features :

- Attitude of limb : **Extended elbow & pronated forearm.**
- 3 point bony relationship : not affected.
- X-ray : Normal.



Treatment :

- Self limiting.
- If painful $\begin{cases} \rightarrow \text{Flex elbow \& forcefully supinate forearm.} \\ \rightarrow \text{Extend elbow \& pronate forearm.} \end{cases}$

UPPER LIMB TRAUMA : FOREARM, WRIST AND HAND

----- Active space -----

Fractures of Forearm

00:01:01

Forearm bones → Radius : Articulates with carpal bones (wrist).
 → Ulna : Articulates with humerus (Ulnohumeral joint).

- ulna : Static
 - Radius : Rotates around ulna
 - Triangular fibrocartilage complex (TFCC) : Stabilizes DRUJ.
- } Results in pronation & supination.

Both bone fracture of forearm :



Note :

POP position in forearm fractures :

Fracture of → Proximal 1/3rd : Supination
 → middle 1/3rd : mid prone
 → Distal 1/3rd : Pronation

Rx → Closed reduction (CR) + POP
 → Open reduction (OR) + IF with plates & screws (Best)

Monteggia fracture :

Fracture of upper 1/3rd ulna (Fracture of necessity)

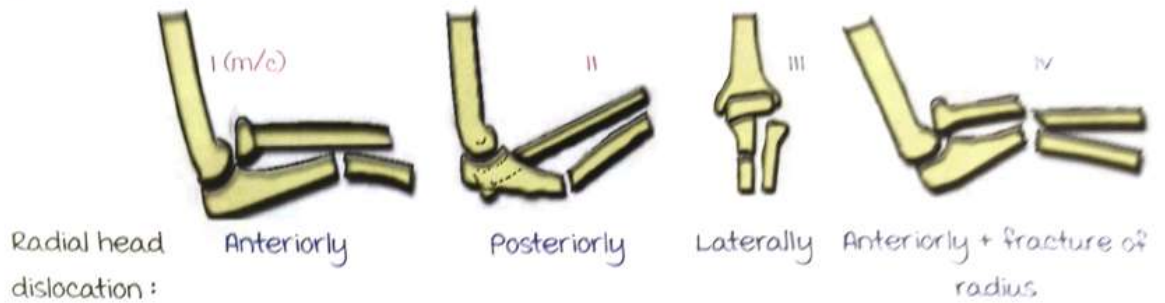
+

Proximal RUJ disruption (Radial head dislocation).



monteggia fracture

----- Active space ----- BADO classification



Treatment : OR + IF with plates (On ulna) & screws $\pm$ K-wires to stabilise PRU.

Complications :

m/c injured nerve : **Posterior interosseous nerve** (Branch of radial nerve)

↓
Causes finger & thumb drop without any sensory deficit.

Note :

m/c injured nerve in
 → Shoulder dislocation : **Axillary nerve**.
 → Supracondylar humerus fracture :
 Anterior interosseous > median nerve.



Monteggia fracture

Galeazzi/Reverse Monteggia fracture/Pied mont fracture :

Fracture of radius + DRUJ disruption/TFCC injury.

Rx : Open reduction (OR) + Internal fixation (IF) with plates & screws.



Galeazzi fracture

Nightstick fracture :

- Isolated *ulna fracture*.
- *Self-defense injury*.

Rx
 ↳ CR + POP application.
 ↳ ORIF with plates & screws.

Note :

a.



Monteggia fracture type I



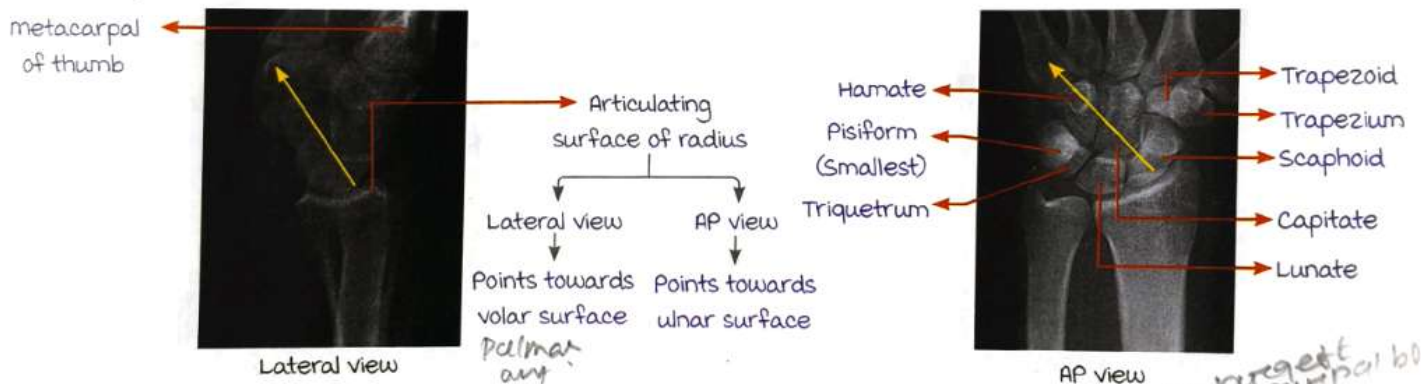
Nightstick fracture

----- Active space -----

b. Nerve injured in posterior dislocation of elbow → Ulnar nerve.

Fractures of Distal End of Radius

00:10:03

Anatomy of wrist :**Fractures at wrist :**

Name	Fracture	Distal fragment displacement
Colles's	Extraarticular	Dorsally
Smith		Ventrally
Barton	Intraarticular	Dorsal/ventral + carpal subluxation
Chauffeur		Isolated radial styloid

capitate - biggest carpal bone
 take 2 months to ossify



Colles fracture



Smith's fracture



Bartons fracture



Chauffeur's fracture

----- Active space -----

COLLE'S FRACTURE

- m/c fracture of distal end of radius.
- Fracture of distal end of radius at cortico-cancellous junction.
- Extra-articular fracture.
- Common in elderly **post-menopausal females**.
- mechanism of injury : Fall on an outstretched hand with wrist in extension.



Colle's fracture

Displacements

1. Dorsal tilt/shift
 2. Lateral tilt/shift
- } Dinner fork deformity.



Dinner fork deformity

Treatment

1. Closed reduction (CR) (Sequence : Traction → Palmar flexion → Ulnar deviation) + POP application (Position : Hand shaking position).
- 2.



Colle's cast

Note :

Colle's cast : **Below elbow**

↓
To avoid stiffness of joint
(m/c complication)

3. Definitive : ORIF with plates & screws.

Complications

- Stiffness of fingers (m/c).
- **malunion** → Dinner fork deformity (2nd m/c).
- **Sudeck's dystrophy**/Complex regional pain syndrome/Reflex sympathetic dystrophy fracture → Nerve injury → Autonomic nervous system mismatch
↓
Tense swelling, shiny skin,
tingling & numbness, paresthesia.
- Rupture of extensor pollicis longus tendon.
- Carpal tunnel syndrome : Fracture haematoma → median nerve compression.
- Non-union (Rare).

SMITH FRACTURE

- Extra-articular fracture of distal end of radius with volar displacement.

↓
Garden spade deformity.

- mechanism of injury : Fall on an outstretched hand with wrist in flexion.



Garden spade deformity

BARTON'S FRACTURE

- Intra-articular fracture of distal radius + Carpal subluxation.
- Rx : ORIF with plate & screws.



Barton's fracture



Chauffeur's fracture

CHAUFFEUR'S FRACTURE/DRIVER'S FRACTURE

- Isolated radial styloid fracture.
- Intra articular fracture.

Fractures of Hand

00:31:25

Carpal bones

'She Looks Too Pretty, Try To Catch Her'

- Scaphoid (S).
- Lunate (L).
- Triquetrum (T).
- Pisiform (P) → Smallest.
- Trapezium (Tz).
- Trapezoid (Td).
- Capitate (C) → Largest.
- Hamate (H).



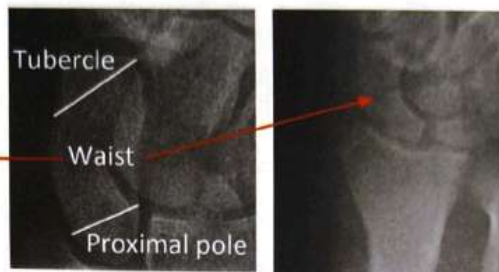
Carpal bones

- Carpal bone to ossify
 - First : Capitate (At 2 months of age).
 - Last : Pisiform.
- m/c carpal bone to
 - Fracture : Scaphoid.
 - Dislocate : Lunate → Causes median nerve injury.

----- Active space -----

SCAPHOID FRACTURE

m/c site of fracture



mechanism of injury : Fall on an outstretched hand.

Age group : Young adults.

Clinical features :Pain/swelling in *anatomical snuff box*.**Evaluation :**

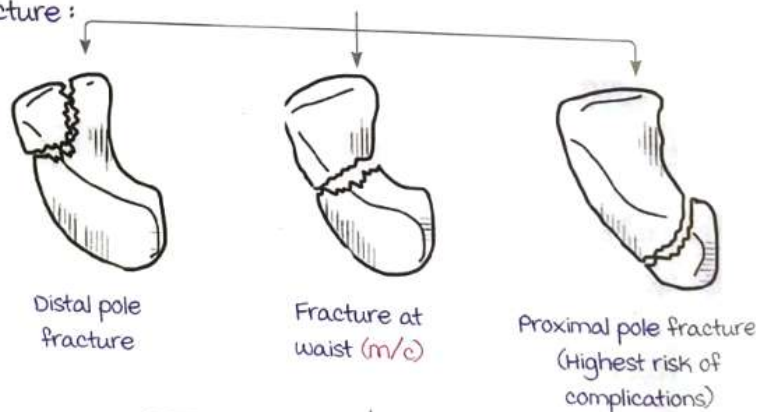
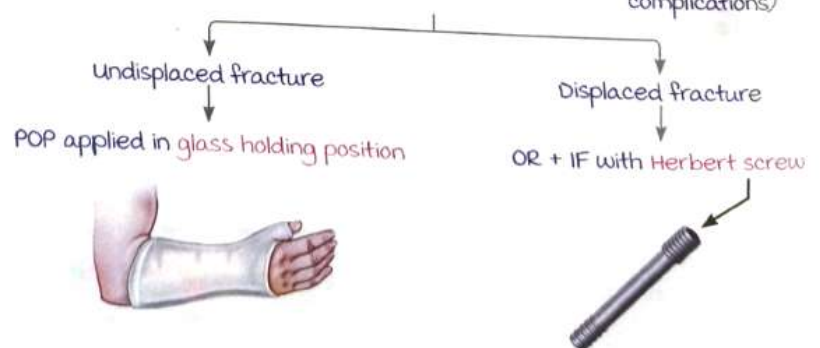
- X-ray
 - Oblique view (*Best*)
 - AP
 - Lateral
 } Fracture may not be visible.
- MRI : For fractures not visible on X-ray.

Complications :

- Non-union (*m/c*).
- Avascular necrosis of proximal fragment.

Types :

Based on location of fracture :

**Treatment :**

Scapho-Lunate ligament injury :

Ligament injury → Scapho-lunate dissociation : **Terry Thomas sign.**
 (↑ gap between the bones : >3-4 mm)

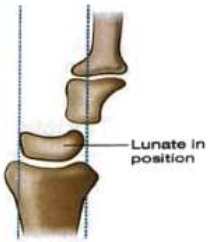
Note :

- Normal gap between scaphoid & lunate : 2-3 mm.
- **Keinbock's disease** : Osteonecrosis of lunate.



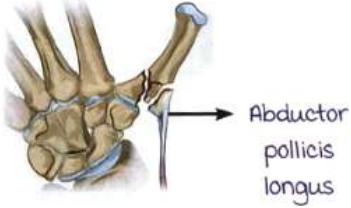
Scapho-lunate ligament

Carpal instabilities :

	Lunate dislocation	Peri-lunate dislocation (mv/c)
Type of dislocation		
Additional features	Spilled coffee-cup sign	Rx : Reduction with Tavernier's maneuver

FRACTURE OF BASE OF FIRST METACARPAL

- Intra-articular fracture of trapezio-metacarpal joint.
- Displacing forces : Abductor pollicis longus pull.

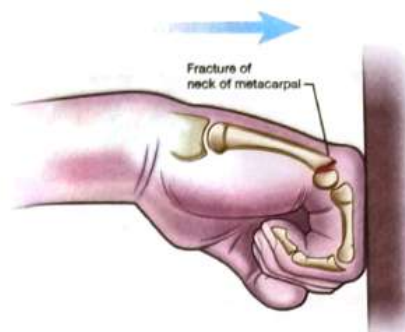
Bennet's fracture	Rolando's fracture
	
Partial	Complete comminuted (T/Y shaped)
Displaced fracture	Undisplaced fracture
	

----- Active space -----

BOXERS FRACTURE

- Fracture of 5th metacarpal.
- Extraarticular.

Rx : CR + POP application → In Jame's position
(For safe immobilization).

**MALLET FINGER V/S JERSEY FINGER**

	mallet finger	jersey finger
mechanism of injury	Hyperflexion injury (Eg : When ball hits the finger) ↓ Injury to extensor digitorum communis tendon (Tear/avulsed) ↓ Flexion of distal phalanx	Hyperextension injury ↓ Injury to flexor tendon (Tear/avulsed) ↓ Extension at DIP
On examination	• Extension at DIP → Passively (+) → Actively (-) • Bony fragment on dorsal side (+)	• Flexion at DIP → Passively (+) → Actively (-) • Bony fragment on volar side (+)
Treatment	mallet/stax/frog splint	-

LOWER LIMB TRAUMA : PELVIS AND TESTS

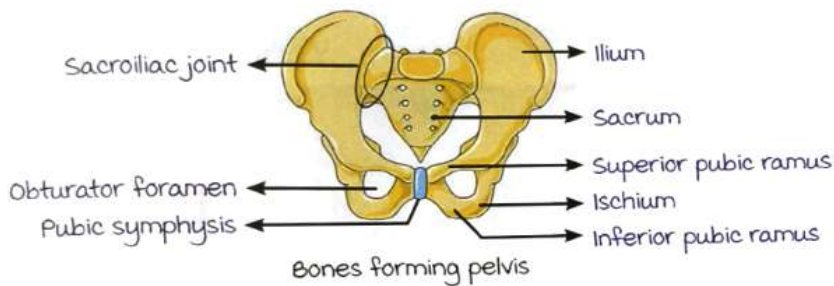
----- Active space -----

Anatomy of Hip Joint

00:00:59

Hip joint is formed by proximal femur & acetabulum of the pelvis.

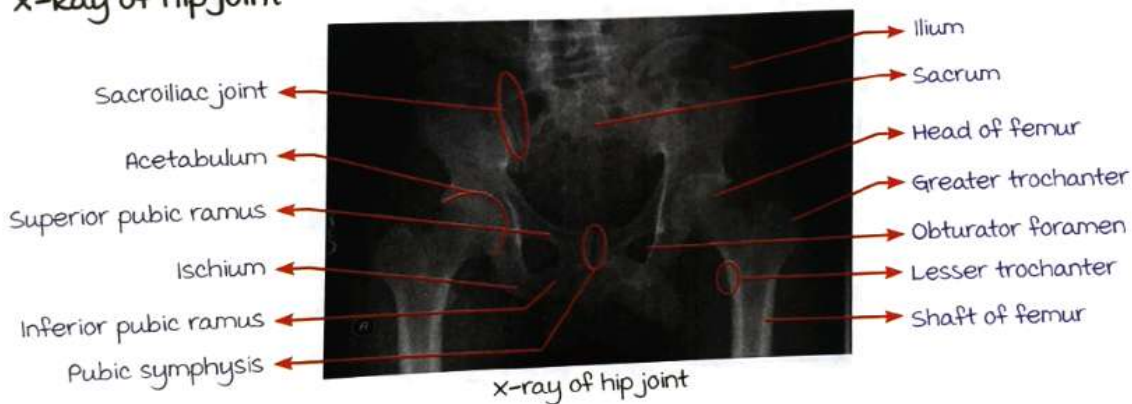
Pelvis :



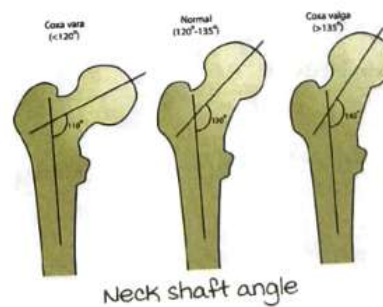
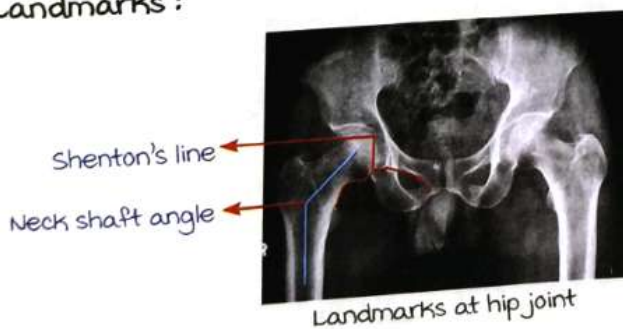
Proximal femur :



X-Ray of hip joint



Landmarks :



Neck shaft angle

1. Shenton's line :

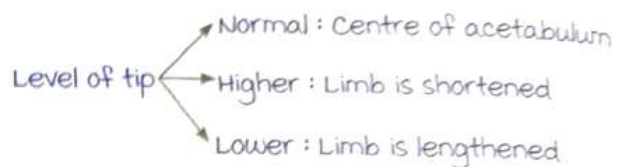
- Continuous line from lower border of superior pubic ramus, laterally towards head and neck of femur.
- Disturbed in any pathology of the hip.

----- Active space -----

2. Neck shaft angle :

- Normally 120° – 135° .
- $<120^{\circ}$: Coxa vara.
- $>135^{\circ}$: Coxa valga.

3. Greater trochanter :



4. Lesser trochanter : Gives an idea about rotation of limb.

- Disappears on X-ray : Internal rotation of limb.
- Prominent on X-ray : External rotation of limb.

Disruption of shenton's line :

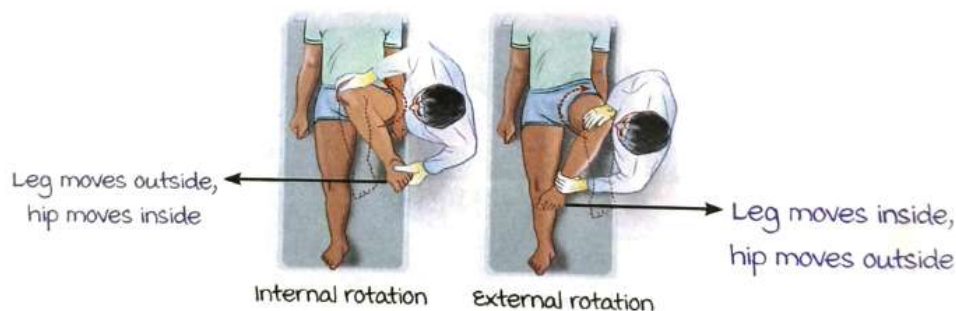


Dislocation of head of femur



Fracture of neck of femur

movements at the hip :



Pelvis Fracture

00:11:03

Pelvis fractures are high velocity injuries.

Jumpers fractures :

- U or H-shaped fracture.
- Etiology : Fall from height.



U shaped fracture



H shaped fracture

TILES CLASSIFICATION (PELVIC FRACTURES)

- Stable : Injury other than the pelvic rim.
- Unstable : Injury to the pelvic rim/ring.

Stable fracture :

Isolated fractures of :

- Ilium.
- Ischium.
- Anterior superior iliac spine (ASIS).
- Anterior inferior iliac spine (AIIS).

Eg : **Duverney's fracture**

Isolated fracture of iliac wing.

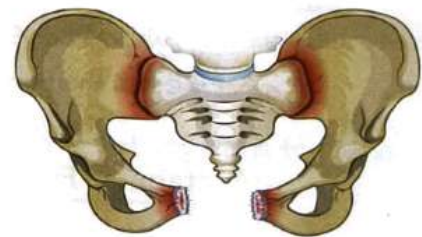
Unstable fracture :

Pelvic ring injured.

Young and Burgess classification : Based on forces causing injury.

1. Anteroposterior compression :

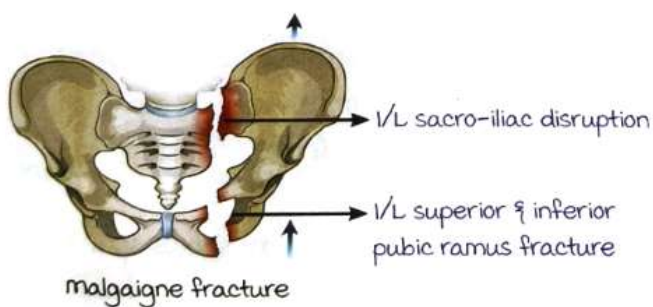
- Force from the front in anteroposterior direction → Pelvis opens up → **Open book fracture**.
- Horizontally unstable fractures.



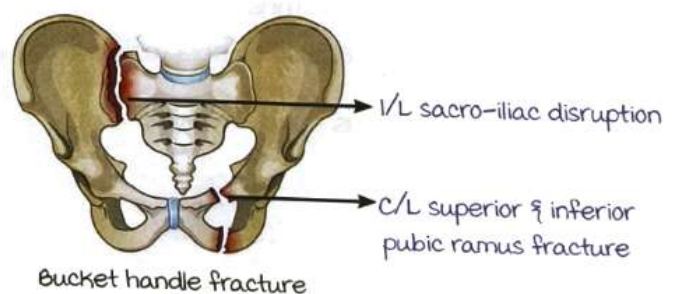
Open book fracture

2. Lateral compression

- Force from the side → Lateral compression of pelvis.
- Pelvis becomes vertically unstable.
- Eg : **malgaigne fracture** and **Bucket handle fracture**.



malgaigne fracture



Bucket handle fracture

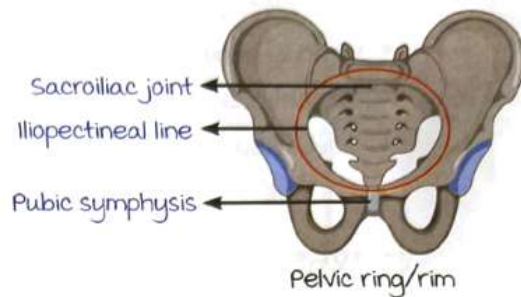
3. Complex force :

- Bilateral superior and inferior pubic ramus fractures.
- Eg : **Straddle fracture**.



Straddle fracture

----- Active space -----



Duverney's fracture

----- Active space -----

Consequences of pelvic fractures

massive bleeding (1.5 - 2L) due to :

- venous plexus around pelvis.
- Cancellous bone.
- Soft tissues around pelvis.
 - urethral injury (m/c).

EMERGENCY MANAGEMENT

Pelvic fractures d/t polytrauma are managed by ATLS (Advanced trauma life support).

- A : Airway.
- B : Breathing.
- C : Circulation, stop the bleeding and resuscitate.
- D : Disability evaluation.
- E : Exposure control.

1. management of bleeding :

- External bleeding : Compression.
- Internal bleeding : Pelvic tamponade.
 - Pelvic binders.
 - Bed sheets/towels.

2. IV fluids : RL/NS to restore the volume.

3. External fixation : In compression mode to maintain tamponade.

Acetabular Fractures

00:23:06

Parts of acetabulum :

- 2 columns.
- 2 walls.

X-Rays : Judet views.

Classification of acetabular fractures :

Judet classification

- Anterior wall.
- Anterior column.
- Posterior wall (m/c).
- Posterior column.
- Bi-columnar : X-Ray shows spur sign.

Lower Limb Tests

00:24:43

----- Active space -----

Trendelenburg test :

- To assess **abductor mechanism** of hip.
- Principle abductors of hip :
 - Gluteus medius
 - Gluteus minimus
 Supplied by superior gluteal nerve.
- Abductors help maintain gait : I/L abductors help swing C/L limb.
- Helps moving the limb when other limb is in stance phase.

Abductor failure :

Causes of abduction failure :

- Gluteus medius weakness.
- Gluteus minimus weakness.
- Superior gluteal nerve palsy.
- Coxa vara.

mnemonic : **SSS**Sound Side Sinks
(Normal side sinks)

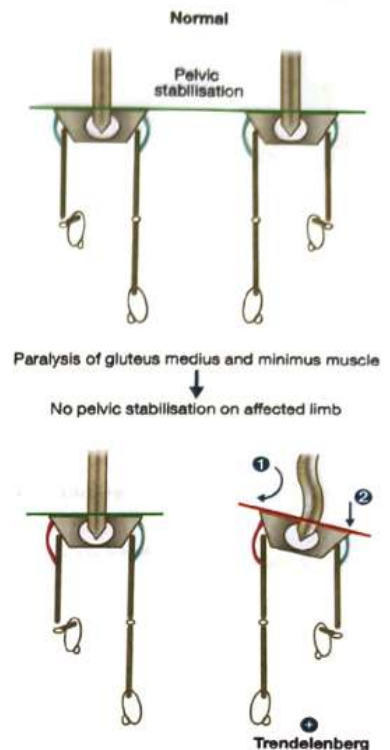
Test procedure :

Ask patient to stand on each limb for 30 sec, to assess its integrity

↓
Observe the ASIS on both sides

↓
When patient stands on the **pathological side**, **Sound side sinks down** → Patient leans to pathological side.

Positive test : ASIS/PSIS of other side goes down.



- 1 Sign positive when made to stand on affected limb
- 2 Pelvic sags on healthy side

Trendelenburg test

Trendelenburg gait :

Patient walks by leaning the spine towards the pathological side to maintain balance.

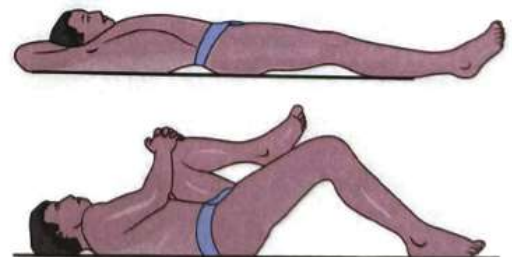
Hand to Knee gait :

Seen in **quadriceps weakness** due to poliomyelitis.

Thomas test :

AKA Hugh Owen Thomas well leg raise test.

Used to assess **flexion contracture/flexion deformity** of the hip.



Thomas test

----- Active space ----- Steps :

Patient lies supine
 ↓
 Observe for exaggerated lumbar lordosis
 ↓
 Flex the unaffected hip the lordosis is obliterated
 ↓
 Look for flexion of other hip (Pathological side)
 ↓
 measure angle b/w the flat table & femur (goniometer)

Thompson test :

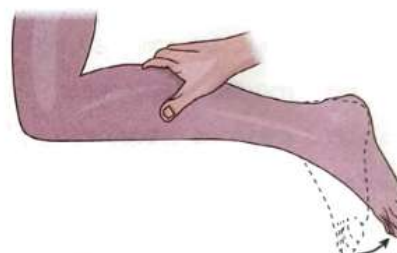
Assess integrity of **tendo achilles**.

AKA Simmonds Thompson test.

Tendoachilles :

- Common tendon of gastrosoleus that inserts on calcaneum.
- Action : **Plantar flexion**.

+ve Thompson test : Failure of plantar flexion on squeezing the calf d/t tendo achilles damage.



Squeezing calf produces plantar flexion of foot when heel cord is intact



Squeezing calf produces no motion of foot in injured leg

Thompson test

LOWER LIMB TRAUMA : HIP DISLOCATION AND FRACTURES

----- Active space -----

Hip joint :

Stable ball and socket joint.

Stability provided by :

- Concave acetabulum that accommodates the femoral head
- Ligaments : Iliofemoral ligament (strongest ligament of the body)
- ↓ chances of recurrence after reduction of dislocation

Hip dislocations

00:00:53

TYPES OF DISLOCATION

Pure dislocation

Head of femur comes out of acetabulum without any other associated injury.

	Posterior dislocation (m/c)	Anterior dislocation
mechanism of injury	Dashboard injury	Deceleration injury Fall from height
Attitude of limb	F, AD, IR : 1. Internal rotation 2. Thigh is adducted, knee is on the other limb   medial side of foot touches the bed d/t internal rotation.	 F, AD, IR : 1. Flexion at hip 2. Abduction at thigh 3. Externally rotated limb (Lateral part of foot touches the bed)

----- Active space -----

Length of limb	Shortening	Lengthening
X-Ray features	 Posterior dislocation Shenton's line is broken Internal rotation of limb Lesser trochanter is not visible	 Head lies outside acetabulum Shenton's line is broken Abduction and External rotation of limb
Palpation	Head palpable in gluteal region	Head palpable in femoral triangle region

Fracture dislocation

Dislocation a/w fractures (acetabulum, head/neck of femur).

	Posterior fracture dislocation	Anterior fracture dislocation	Central fracture dislocation
Palpation of head of femur	Gluteal region	Femoral triangle	On per rectal examination



Head has breached into the acetabulum and is in the pelvic cavity

Central fracture dislocation

MANAGEMENT

Hip dislocations are an emergency: ↑ risk of avascular necrosis.

Closed reduction

↓ not reducing d/t muscle spasm

Closed reduction under anesthesia

↓ still not reducing

Open reduction

↓

Apply skeletal traction

Note: Shoulder reduction by modified Kocher's technique.

Examples:

- Allis manoeuvre
- Bigelow manoeuvre
- Stimson manoeuvre
- East Baltimore lift manoeuvre

Complications :

1. **Avascular injuries** : m/c (in 6-12 hrs).
2. **Sciatic nerve injury** : In post dislocation.
Presents with foot drop.
3. **Femoral vessel injury** > Femoral nerve injury : In anterior dislocation.

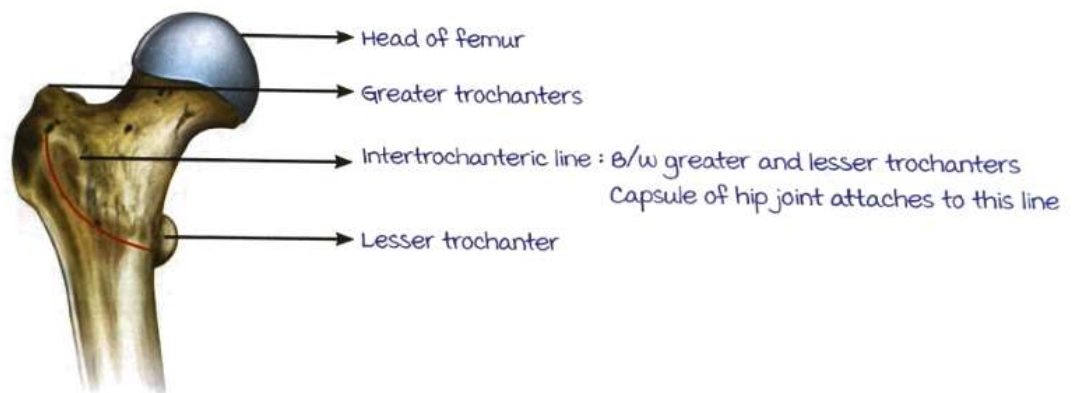
----- Active space -----

Note : vascular sign of narah

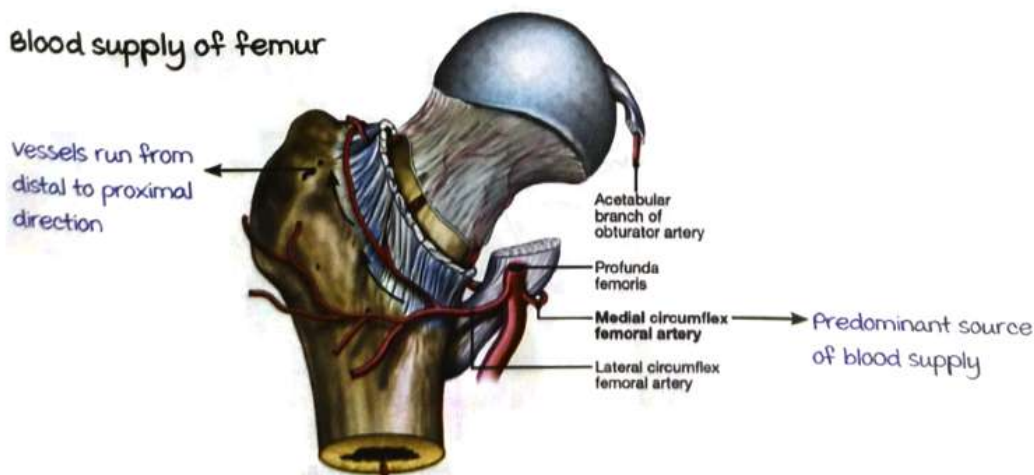
- Inability to palpate the femoral artery against the femoral head while popliteal, dorsalis pedis and posterior tibial artery are palpable.
- Seen when femoral head is not in its original place.
 - Hip dislocation.
 - Congenital dysplasia of hip.
 - Excision of femoral head.

Proximal femur fractures

00:17:04

ANATOMY

Anatomy of proximal femur

Blood supply of femur

----- Active space ----- TYPES

1. Neck of femur fracture : Intracapsular (Proximal to intertrochanteric line).
2. intertrochanteric fracture : Extracapsular (Distal to intertrochanteric line).

	Neck of femur/Intracapsular fracture	Intertrochanteric/extracapsular
Age (elderly)	50-60 yrs	50-60 yrs
Sex	Female >>> males	Female > males
Trauma	Trivial fall	moderate to severe fall
Pain	mild pain	moderate to severe pain
Location of pain	Scarpa's triangle	Trochanteric region
Shortening	< 1 inch	> 1 inch
Deformity/attitude	External rotation <45° (Capsule limits it)	External rotation >45° (Lateral part of foot touches the bed)
Complication	AVN (45°) > Non-union (30%) Due to disruption of blood supply	malunion → Coxa vara Does not disrupt blood supply
X-Ray	 Intracapsular fracture Shenton's line Joint capsule	 Extracapsular fracture Joint capsule

Non-union : Seen in neck of femur fracture

Causes :

- Synovial fluid washes the fracture hematoma } Prevents healing
- ↓ vascularity
- Neck of femur lacks cambium layer that has osteoprogenitor osteoblasts

NECK OF FEMUR FRACTURE :**Classification :**

Anatomical classification : Based on location

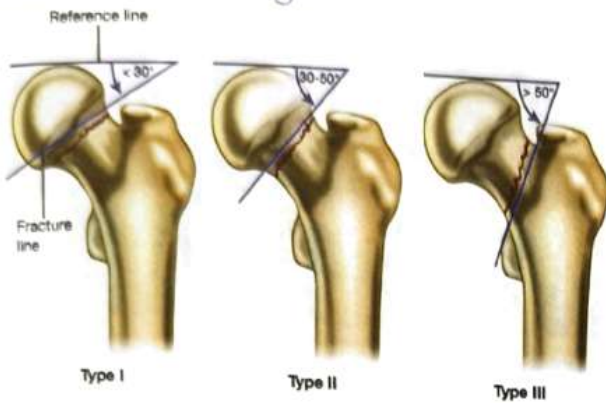


Subcapital fractures : **Highest** chances of avascular necrosis and non-union

----- Active space -----

Pauwels' classification of transcervical fractures :

Based on Pauwels' angle



Significance : more vertical the angle

Greater is the displacement

Poor prognosis

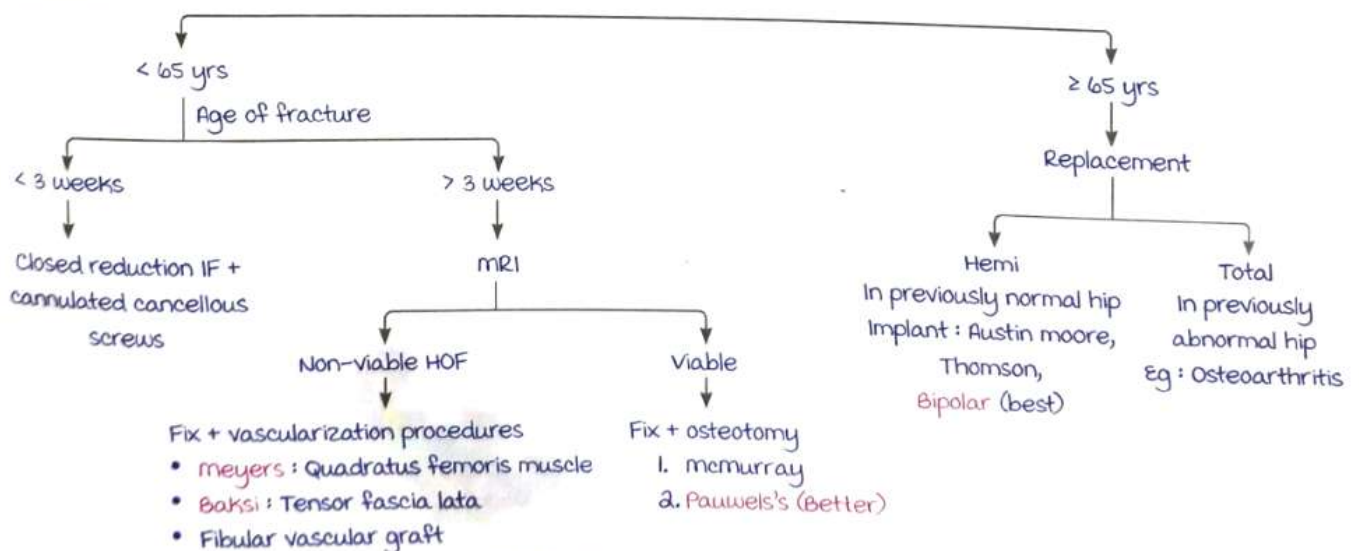
Prognosis : Pauwels' I > II > III

Garden's classification :

	Stage I	Stage II	Stage III	Stage IV
Images	 Trabeculae bending only Valgus impacted	 Trabeculae between head and neck broken and displaced	 All three sets of trabeculae out of alignment	 Acetabular and head trabeculae in alignment, between head and neck not aligned
Type	Incomplete	Complete	Complete	Complete
Displacement	undisplaced	undisplaced	Partially displaced	Completely displaced

Prognosis : I > II > III > IV

Treatment :



----- Active space -----

Hemi replacement : Replacement of only head and neck of femur.

Total replacement : Replacement of head & neck of femur + acetabular cup.

CRIF with cannulated
cancellous screws

Hemi replacement



Total replacement

Prosthesis :



Bipolar



Austin moore



Thompson

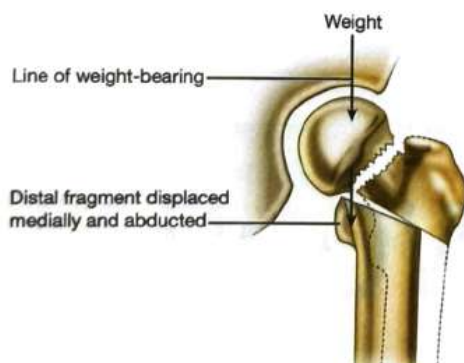
Vascular procedures :



meyer's muscle graft

Osteotomy :

1. mcmurray's osteotomy



- Cut the bone such that shaft is medialised
- weight is transmitted from HOF → shaft bypassing fracture line

2. Pauwells's osteotomy

- Converts vertical fracture pattern to horizontal pattern
↓
change in fracture orientation
- when patient puts weight → Impaction + healing

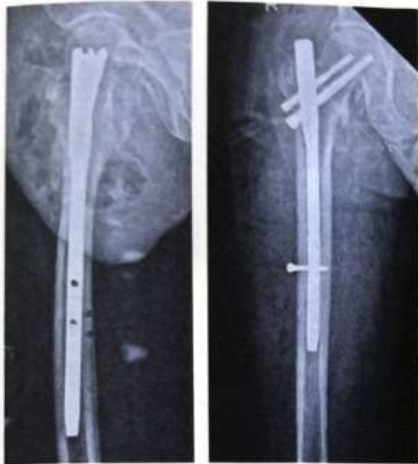
Note : In patients with H/O fall and hip pain with no abnormality on X-ray

MRI to rule out occult neck of femur fracture

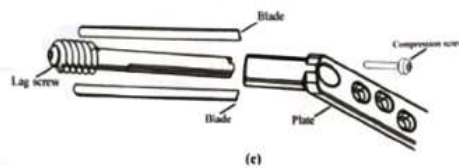
INTERTROCHANTERIC FRACTURE management :

----- Active space -----

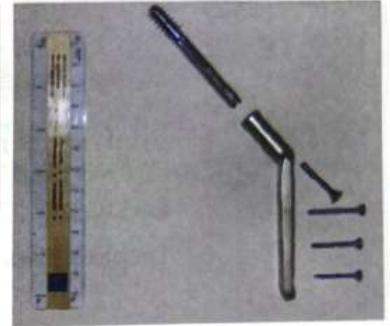
1. Surgical : maintain neck shaft angle with devices & prevent Coxa Vara.



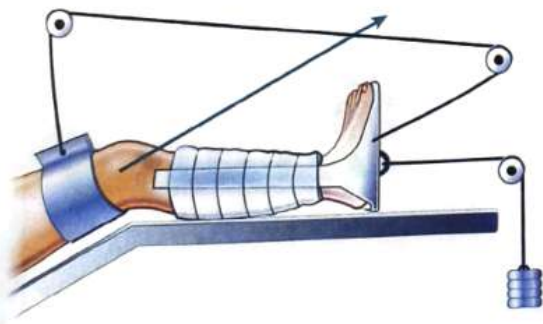
a. Proximal femoral nail with locking and stabilization screws



b. Dynamic hip screw



2. Conservative management :



a. Russell's traction



b. Derotation boot

----- Active space -----

LOWER LIMB TRAUMA : FEMUR, KNEE, LEG AND FOOT

Shaft of Femur Fracture

00:00:06

BASICS

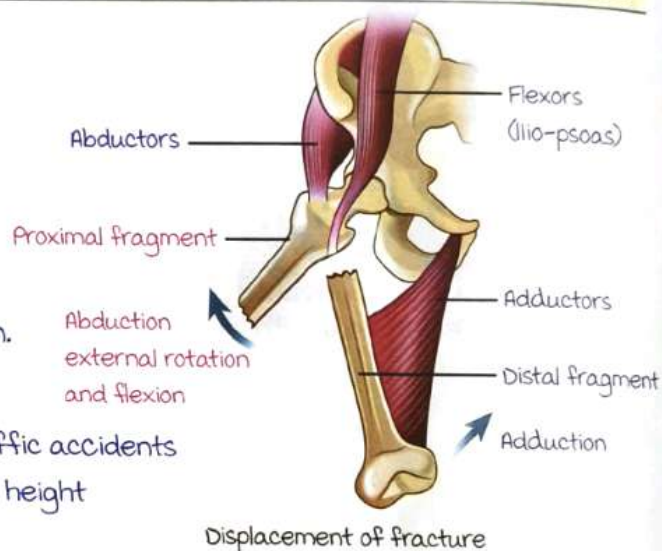
Displacement of fragments :

Classification :

Winkler and Hansen classification.

Mechanism of Injury :

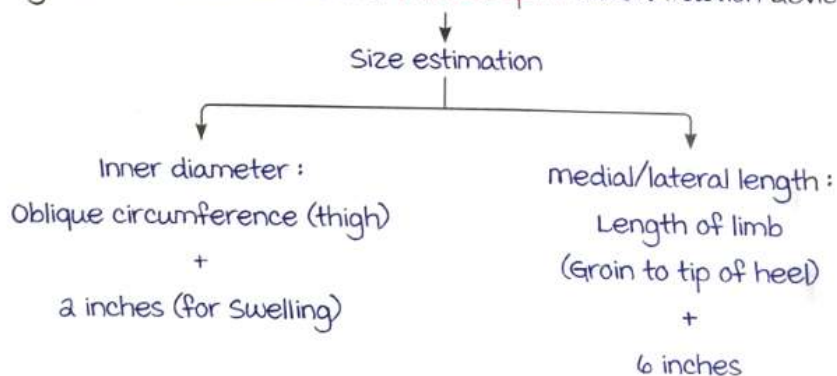
High velocity injuries → Road traffic accidents
→ Fall from height



MANAGEMENT

mx in adults

Temporary fixation : Stabilisation with Thomas splint (fixed traction device).



Definitive management : CRIF with intramedullary nail with Interlocking screws.

Complications :

1. Stiffness of knee joint.
2. Delayed/non-union.
3. Hypovolemic shock (1 - 1.5L blood loss).
4. Fat embolism syndrome.

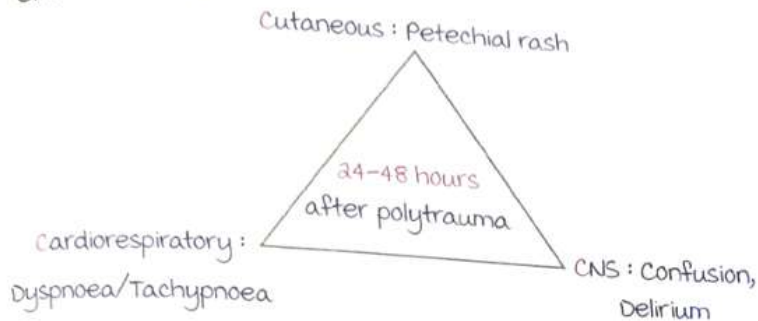


Intramedullary Rod

Fat Embolism

Pathogenesis : Leaking of intramedullary fat into circulation.

Clinical features :



Note :

- Pain on passive stretch → Compartment Syndrome.
- H/o massage → myositis ossificans.

D/d : Acute respiratory distress syndrome (ARDS)

GURDS Criteria :

major Criteria (4)	minor Criteria (8)
• Axillary/Subconjunctival petechia • PaO₂ below 60mmHg • CNS Depression • Pulmonary edema	• Tachycardia • Fever • Anemia • Thrombocytopenia • Fat globules in sputum • Fat globules in urine (Lipuria) : Gurd test • ↑ ESR • Retinal emboli
1 major + 4 minor = Fat embolism	

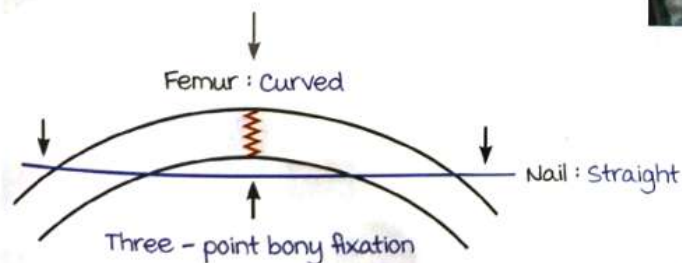
management → Prevention : Immobilize and fix fracture early.
 → Treatment : Damage control orthopaedics
 External fixation → ICU Stabilization → Definitive mx

mx in children

- <6m : Pavlik Harness.
- 6m-5y : Hip spica cast.
- 5-10 y : Flexible nails (Enders nail/Titanium Elastic Nail System, TENS).
- >10y : Intramedullary Interlocking nails.

Note :

- <2yrs/<12kgs : Gallows traction.
- Kuntscher Nail : Isthmus of femur.



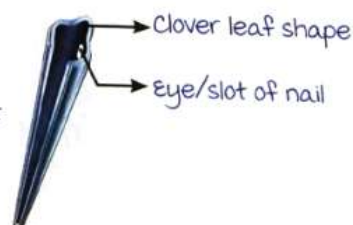
Pavlik Harness



Hip spica cast



Gallows traction



Kuntscher nail



TENS

----- Active space -----

Leg Injuries

00:11:25

X-ray knee joint
lateral view**PATELLA FRACTURE**

Direct trauma to knee.

Transverse fracture of patella → Stimulation of quadriceps tendon → Strong force displacing proximal patellar fragment.

X-ray knee : AP view/lateral view (Skyline view).

management :

- Convert distractive forces into compressive forces → Tension Band wiring (Best) with K-wires.
- Cylinder cast (Tube cast).

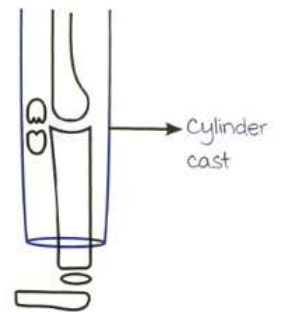
Note :

Olecranon fracture

(Displacement of fragments d/t triceps tendon) : Tension band wiring.



Tension band wiring (TBW)



Cylinder cast

Bipartite Patella :

Congenital Anomaly.

small separated fragment d/t incompletely fused patella at superolateral pole.

Incidental finding on X-ray : $B/L > W/L$.

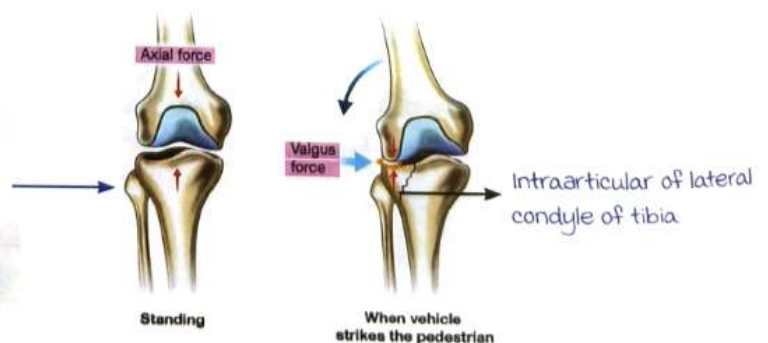
management : conservative.



Bipartite patella.

BUMPER FRACTURE

mechanism of injury : valgus force with Axial loading.



management : Open reduction + internal fixation with plates.

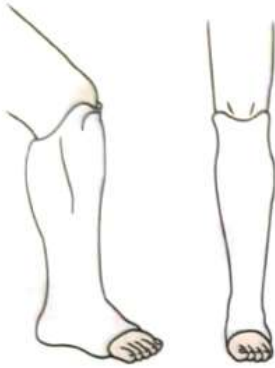
TIBIAL SHAFT FRACTURE

management :

1) Patellar tendon bearing cast

2) Definitive treatment : CRIF with intramedullary rod/nail with interlocking screws

----- Active space -----



Patellar tendon bearing cast

Intramedullary Rod/
Nail with screwsTibia and fibular shaft
fracture

Foot Injuries

00:18:40

Anatomy of Ankle :



AP view



Lateral view



mortis view

Ankle fractures :

Types :

Lateral malleolus



Lateral malleolus fracture

Lateral malleolus medial malleolus



Bimalleolar (Pott's) fracture

Posterior malleolus medial malleolus Lateral malleolus



Trimalleolar (Cotton's) fracture

----- Active space -----

management :

Neurovascular assessment

Closed Reduction

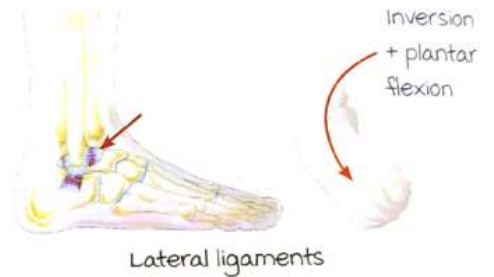
Neurovascular assessment

Slab application

Tibiotalar
dislocationLateral
malleolar

Ankle sprain :

movement	Ligament Injured
Inversion and Plantar Flexion	Anterior Talofibular Ligament (m/c injured in whole body)
Eversion (Rare)	Deltoid Ligament



Pilon/ceiling fracture :

Intra articular comminuted fracture of distal end of tibia.



Calcaneum fracture



mechanism of injury :

Fall from height landing on feet.

Angles :

- ↓ Bohler's angle
- ↑ Gissane's angle



Lover's fracture :

B/L calcaneum Fracture + vertebral compression fracture



----- Active space -----

Aviator's fracture :

- Fracture talar neck.
- ↑ Risk of avascular necrosis of body of talus.
- Hawkins classification of talar neck fracture.

Note :

Hawkins sign : Sign of viability.

Fracture neck of talus



Type I
Undisplaced



Type II
Subtalar dislocation



Type III
Subtalar and tibiotalar
dislocation



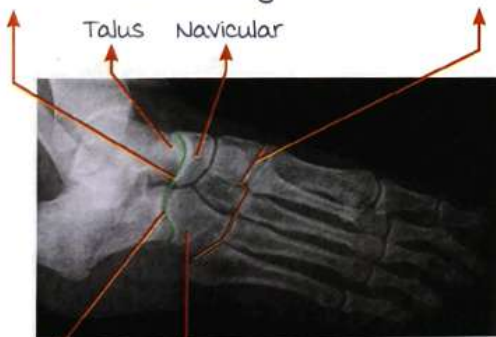
Type IV
Subtalar, tibiotalar, and
talonavicular dislocation



Hawkin's sign

Chopart's fracture :

Fracture of intertarsal joint.



Talus Navicular

Calcaneum Cuboid

Lisfranc's fracture :

Fracture of tarso-metatarsal joint.

Dislocation of second metatarsal from the middle cuneiform



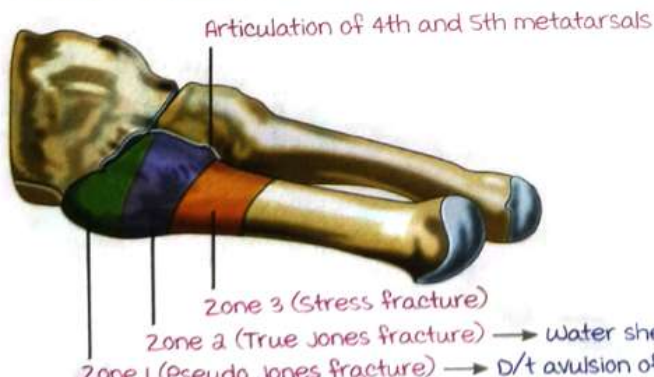
Widened 1st intermetatarsal space

Fracture

Fleck sign : Injury To Lisfranc ligament (Between 1st and 2nd metatarsal)

Robert Jones fracture

Fracture of base of 5th metatarsal.



Articulation of 4th and 5th metatarsals

Zone 3 (Stress fracture)

Zone 2 (True Jones fracture)

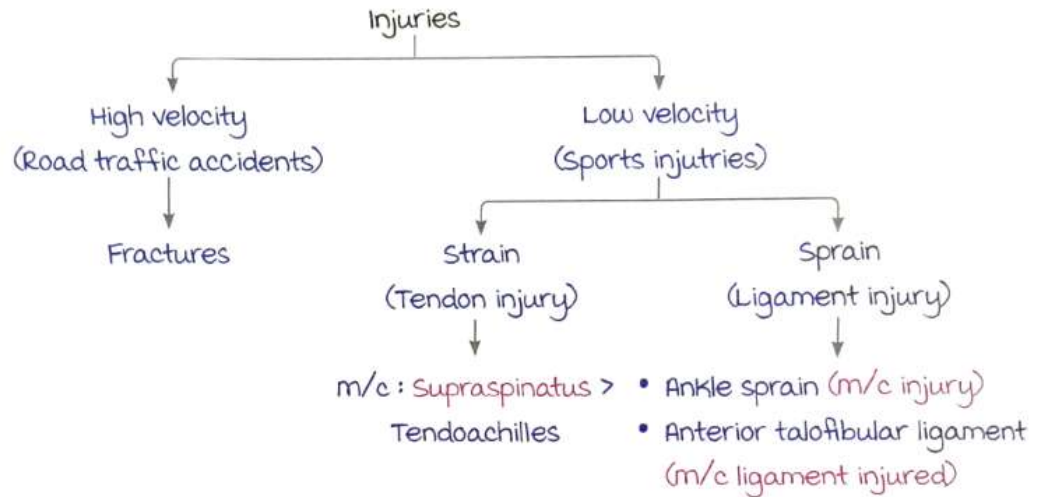
Zone 1 (Pseudo Jones fracture)

→ water shed area (↓ vascularity) → Non-union.

→ D/t avulsion of peroneus brevis tendon.

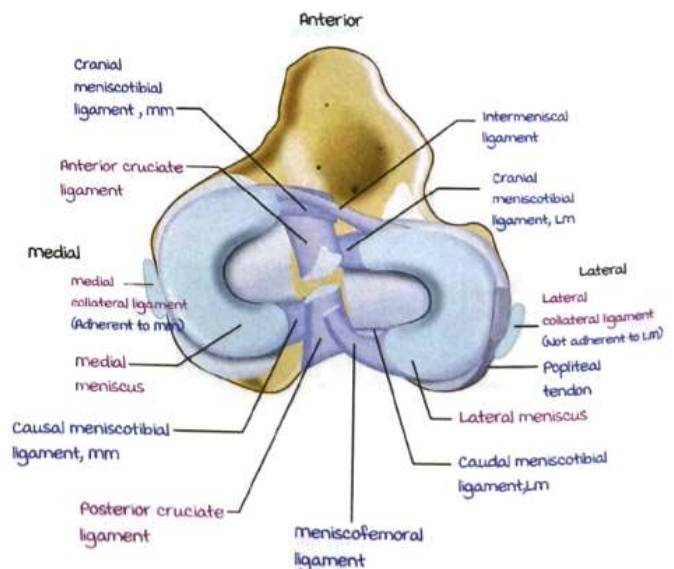
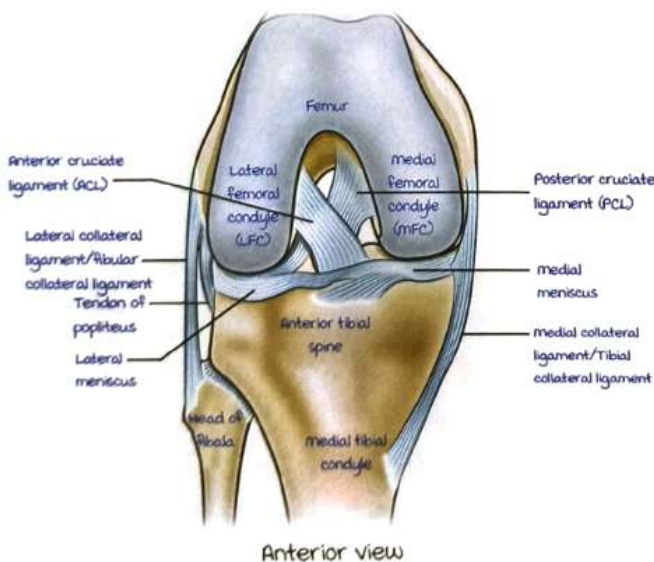
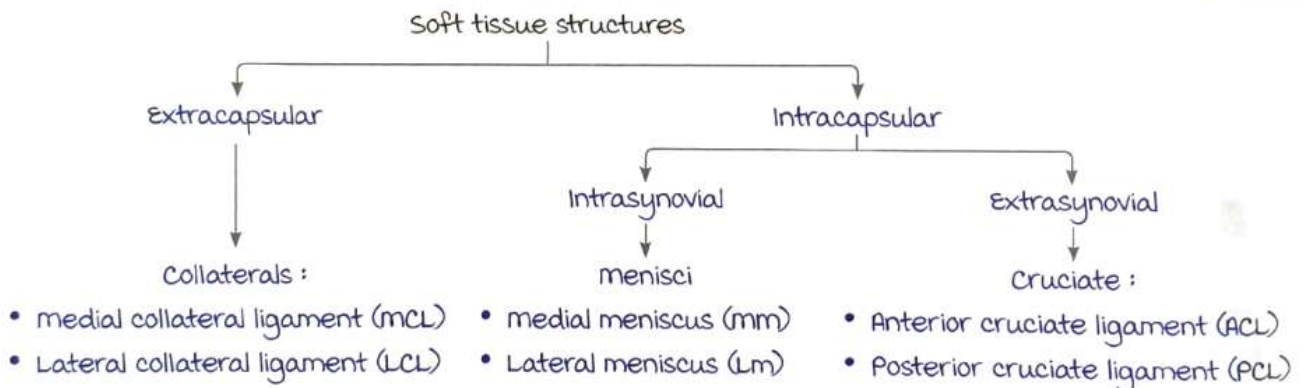
----- Active space -----

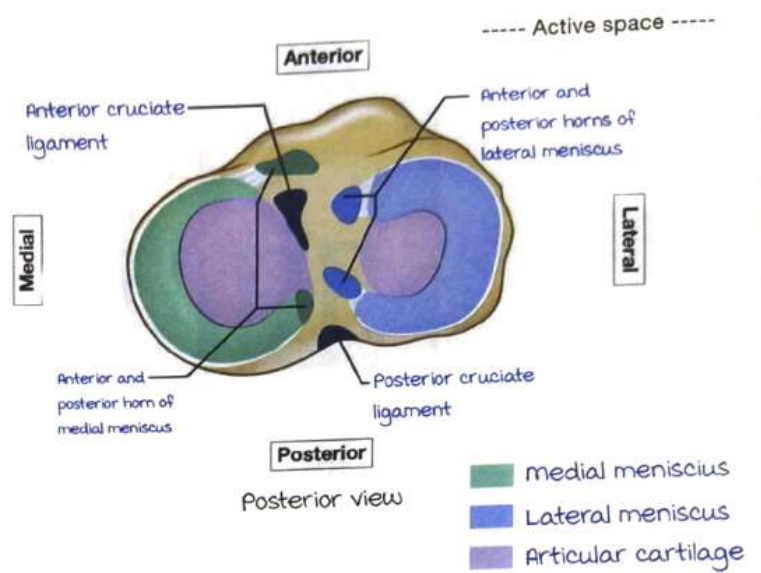
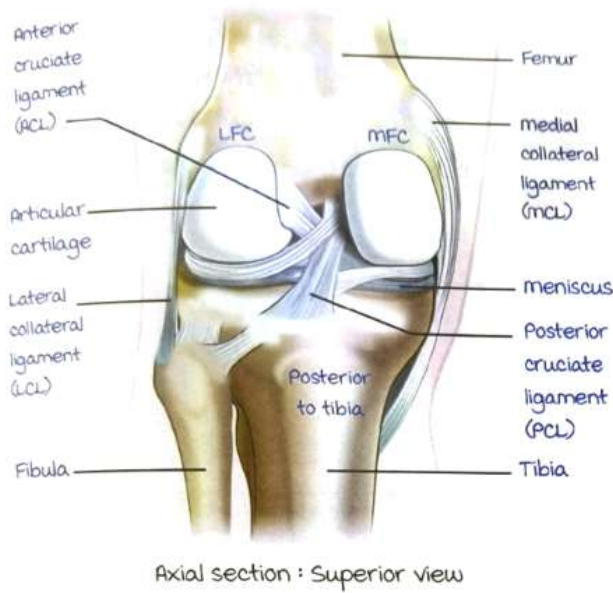
SPORTS INJURIES AROUND KNEE



Anatomy of Knee Joint

00:01:33





Note : For sports injuries

- Gold standard : Arthroscopy.
- IOC : MRI.

Injury to Collaterals

00:08:08

MECHANISM OF INJURY

Function of collaterals :

Coronal plane stability.

Forces :



MCL (m/c injured around knee) > LCL.

CLINICAL FEATURES

	MCL injury	LCL injury
Knee pain	medial side	Lateral side
Examination (Provoke symptoms/open joint space)	At 30° flexion of knee	
	valgus stress test	Varus stress test

----- Active space -----

MANAGEMENTEdema d/t
MCL tear

IOC : MRI (Coronal view)

Treatment

Conservative brace (90%)

Surgery (10%)

Injury to Menisci

00:17:10

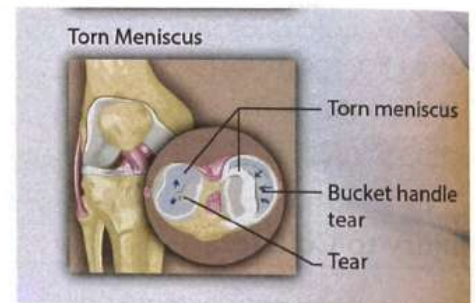
menisci are semilunar structure : Consists of type I collagen.

MECHANISM OF INJURY**Function of menisci :**

1. Shock absorbers.
2. Rotational stabilizers.

Forces

- Torsion/forceful twisting of knee.
- **Bucket handle tear** (m/c type of injury).
- **mm > Lm** injury.



Torn meniscus

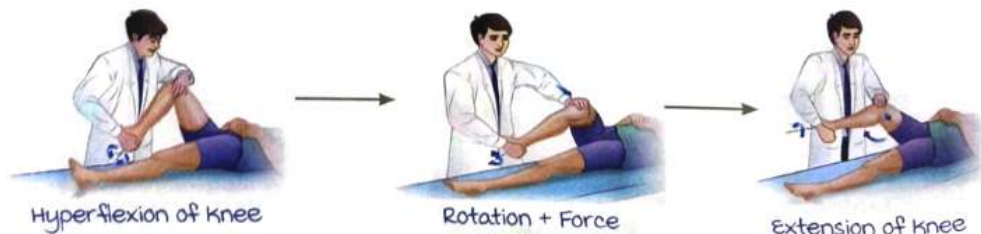
CLINICAL FEATURES**Symptoms :**

Delayed onset.

- Knee pain.
- Swelling : mild to moderate effusion.
- **Pathological locking of knee** (Incomplete extension) : D/t trapping of fragment of menisci between tibial and femoral condyles.

Examination :

To provoke symptoms.

1. **McMurray's test :**

↑ pain/click on :

- Internal rotation + varus force : Lm injury.
- External rotation + valgus force : mm injury.

a. Apley's grinding test :



Prone position → At 90° flexion of knee

a. Rotate menisci between tibial & femoral condyle

+

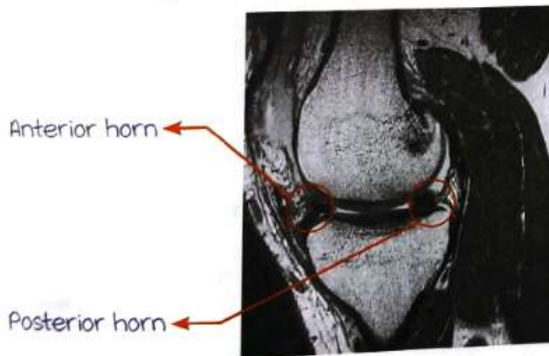
b. Push tibia towards table

3. Joint line tenderness : Best test.

Tender point	Structures injured
Anteromedial	Anterior horn of mm
medial	Body of mm
Posteromedial	Posterior horn of mm
Anterolateral	Anterior horn of Lm
Lateral	Body of Lm
Posterolateral	Posterior horn of Lm

MANAGEMENT

IOC : MRI (Sagittal view).



Normal

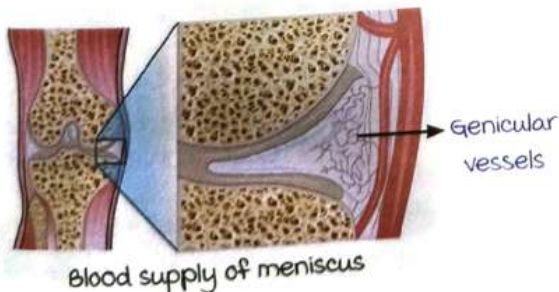


Posterior horn meniscal tear

Heterogenous appearance

Gold standard : Arthroscopy.

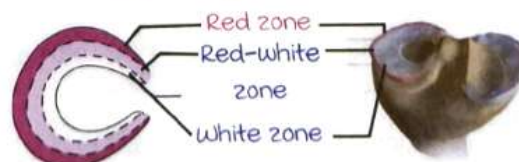
Note :



Blood supply of meniscus

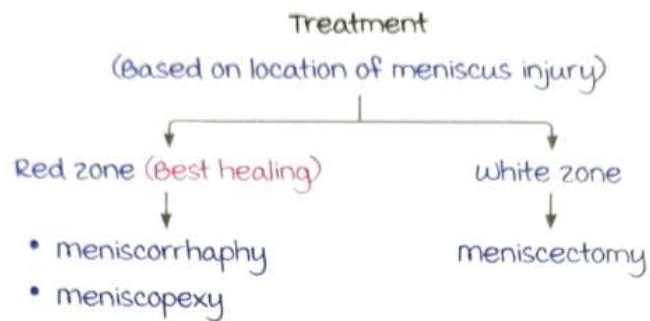
Nutrition to meniscus :

- Peripheral : Genicular vessel.
- Central : Synovial fluid.



Zones based on blood supply

----- Active space -----



Injury to Cruciate

00:29:54

MECHANISM OF INJURY

Function :

Sagittal plane stabilizers.

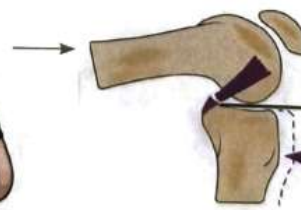
	ACL	PCL
Prevent translation of tibia	Anteriorly	Posteriorly
Knee movement prevented	Hyperextension	Hyperflexion

Forces :

1. Forceful anterior translocation/hyperextension
2. Forceful posterior translocation/hyperflexion



Dashboard injury



CLINICAL FEATURES

Symptoms :

Acute onset

- Twisting injury to knee with a pop → Hemarthrosis (Collection of blood in the joint).
- Instability on walking.

examination :

To test ACL injury

1. **Anterior drawer test :**

- In acute knee : Pain and spasm of muscle inhibits anterior translation.



Anterior drawer test

----- Active space -----

90° flexion of knee
(45° hip flexion)Anterior translation
of tibia for >3 mm2. **Lachmann's test :** most sensitive/best test.

15°-20° flexion of knee

Anterior translation of tibia

3. **Pivot shift test :** most specific

To test PCL injury :

1. **Posterior drawer test :**

90° flexion of knee

Posterior translation/
sag of tibia2. **Godfrey sag test :** Preferred in U/L injury

90° flexion at hip & knee

Posterior translation
d/t gravity**MANAGEMENT**

IOC : MRI.

Gold standard : **Arthroscopy.**

Normal MRI



ACL tear with buckling of PCL

Treatment : **Arthroscopic reconstruction with grafts.**

Grafts :

- Hamstrings : Semitendinosus, Gracilis.
- Patellar tendon.

----- Active space ----- Note :

1. Arthroscopy :

m/c performed on joints : **Knee** > shoulders.

- Anteromedial portal : Instrumentation.
- Anterolateral portal : Arthroscope (4mm/30° scope).



Arthroscopy

2. ACL : 2 bundles :

	Posterolateral bundle	Anteromedial bundle
maintains stability in	Extension	Flexion
ACL tear	Involved in partial/complete tear	Involved in complete tear
Test to assess stability	Lachmann's test	Anterior drawer test

3. Second fracture

- Capsular avulsion in proximal lateral aspect of tibia.
- Associated with ACL tear.



Second fracture

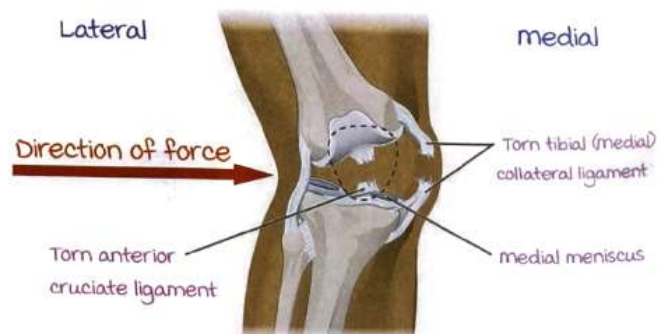
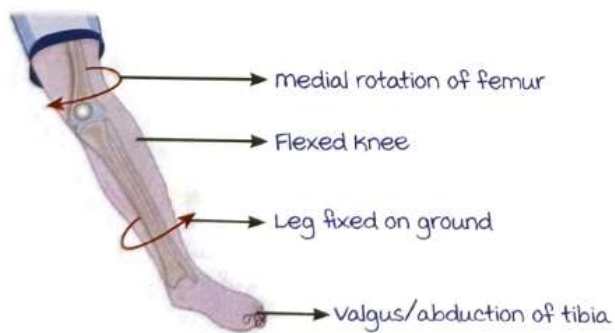
Avulsion of proximal lateral aspect of tibia

Rotational Injuries

00:58:07

TWISTING INJURY TO KNEE

mechanism of injury :



unhappy/painful/terrible/o' Donoghue triad



Sagittal plane : ACL tear



Coronal plane : MCL tear



Sagittal plane : medial meniscus injury

O'Donoghue triad on MRI

----- Active space -----

Note :

Insall-Salvati Index : To assess patellar instability.

- Tendon length (TL)/Patellar length (PL) = 0.8-1.2
- $\uparrow$ TL (>1.2) : Patella alta (Recurrent patellar dislocation).
- $\downarrow$ TL (<0.8) : Patella baja.



Lateral X-ray knee

INJURY DUE TO ROTATIONAL INSTABILITY

	Anterolateral rotatory instability	Posterolateral rotatory instability
Structures injured	• ACL • Anterolateral ligament complex	• PCL • Posterolateral corner structures
Test	Pivot shift test	Dial test

----- Active space -----

PAINFUL REGIONAL CONDITIONS AND AVN

Independent conditions with : Repetitive microtrauma → Pain.

Clinically diagnosed : using tests that replicate the movement causing pain.

Conditions of Shoulder

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Presents with shoulder pain.

Absent h/o trauma.

FROZEN SHOULDER (ADHESIVE CAPSULITIS)

- **Periarthritis** of shoulder joint.
- Self limiting condition.
- 30% bilateral.

Associated conditions :

- Diabetes mellitus (Uncontrolled hyperglycemia).
- Thyroid disorders.

Pathophysiology :

Capsulitis (Inflammation of capsule) of shoulder joint → Pain

↓
Thickening & fibrosis

↓
Contracture of capsule

↓
Restriction of **active & passive** movements of shoulder

Stages :

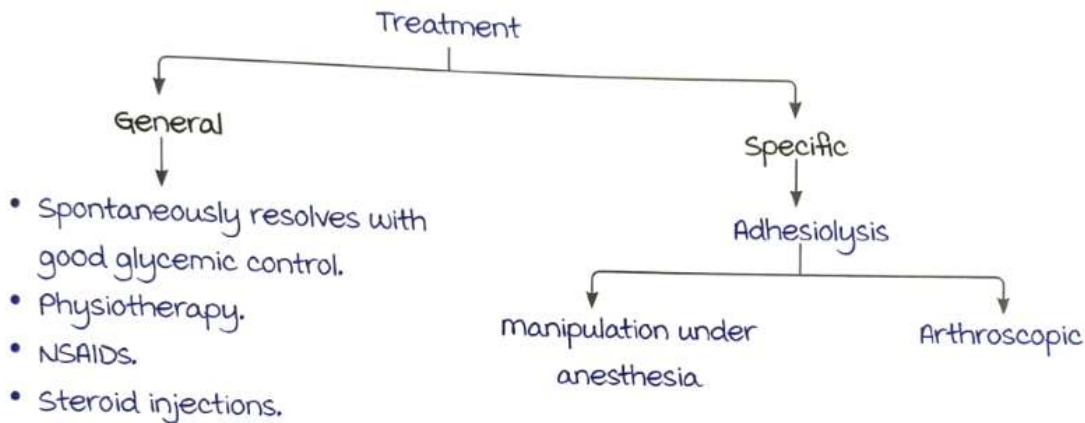
Stage	Pathology	Pain	Stiffness
I : Freezing	Capsulitis	+	-
II : Frozen	Adhesions formed	↓ pain	↑ stiffness
III : Thawing	Resolution	↓ pain	↓ stiffness

Evaluation :**Clinical :**

- Global ROM (Restriction of movements).
- Earliest movement restricted : **Rotation** (External > Internal).

X-ray : Normal.

MRI : Capsulitis (+).

Treatment :**IMPINGEMENT SYNDROME (PAINFUL ARC SYNDROME)****Anatomy of shoulder joint :**

Type of joint : Ball & socket.

Stabilizers : **Soft tissue** > bony.

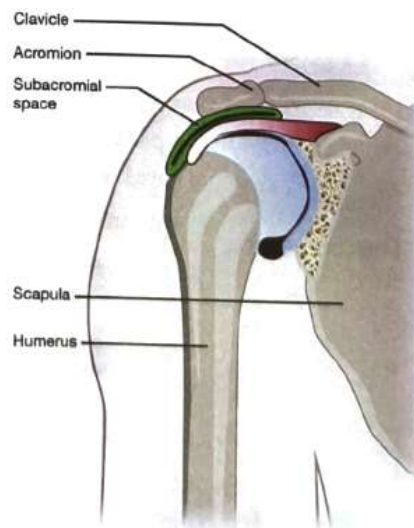
1. Long head of biceps tendon.

2. Rotator cuff muscles.

- Supraspinatus superiorly.
- Subscapularis anteriorly.
- Infraspinatus.
- Teres minor. } Posteriorly

Subacromial space :

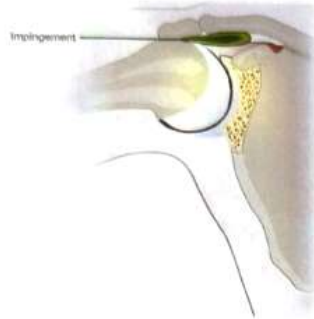
- Space between : Acromion & head of humerus.
- Contents :
 - a. Subacromial bursa.
 - b. Supraspinatus tendon.



----- Active space -----

• Clinical aspect :

In individuals with narrow subacromial space : Shoulder abduction



Shoulder abduction

Inflammation of

Subacromial bursa

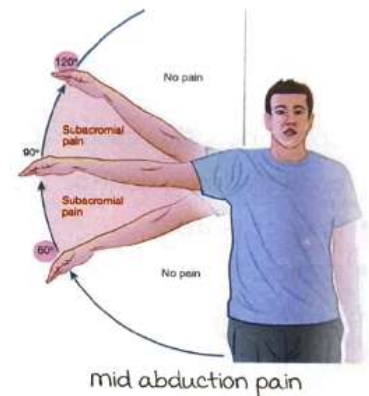
Subacromial bursitis → Supraspinatus tendonitis

Clinical features :

• Shoulder pain :

- Exaggerated on shoulder abduction.
- Maximal pain : mid-abduction

Hence k/a painful arc syndrome



Diagnosis :

Neer's impingement test :



Keeping the scapula stable, the affected arm is raised fully in passive flexion, abduction, & internal rotation

Hawkins-Kennedy impingement test :



The affected arm is elevated 90 degrees in the scapular plane with the elbow in 90 degrees flexion. The upper arm is then stabilized & rotated internally

Treatment :

Conservative :

- NSAIDs.
- Physiotherapy.
- Avoid overhead abduction activities.

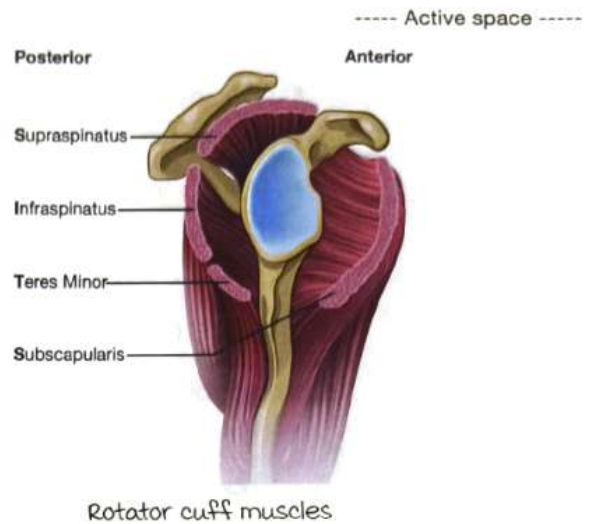
Interventions :

Subacromial decompression

ROTATOR CUFF TEAR & RCT ARTHROPATHY

- m/c tendon to tear: **Supraspinatus** > tendo achilles.
- Forgotten tendon of rotator cuff: **Subscapularis**.

Pathophysiology:



Etiology:

1. Degeneration in elderly.
2. Trauma in young individuals.

Evaluation:

X-ray: Normal.

MRI: **IOC**.

Clinical examination:

Clinical features:

- Pain at the shoulder.
- ↓ ROM of shoulder.
- Pain on performing activity of torn tendon.
- ↑ Pain on movement against resistance.

muscle	Action	Test
Supraspinatus	Abduction	Beer can test, empty can test, Jobe's test
Infraspinatus	External rotation	Horn Blower's test
Teres minor		
Subscapularis	Internal rotation	Lift off test, belly press test, bear hug test



Empty can test

The patient is asked to resist the downward force by the examiner when the arms are in the position of 90° abduction, 30° flexion, and complete internal rotation.



A



B

Tests for subscapularis



Jobe's test

Arms elevated in the scapular plane, elbows extended with the thumb pointing downwards, while the examiner applies downward pressure.

----- Active space -----

Treatment :

- Avoid painful & overhead activities.
- Physiotherapy.
- NSAIDs.
- Steroid injections.
- Arthroscopic tendon repair.

RUPTURE OF LONG HEAD OF BICEPS

Chronic rotator cuff tear → Shoulder instability

↓
Excess stress on other stabilizers of shoulder

↓
Degeneration & rupture of tendon of long head of biceps

↓
muscle belly retracts into distal arm

↓
Popeye sign : Bulge in the distal arm

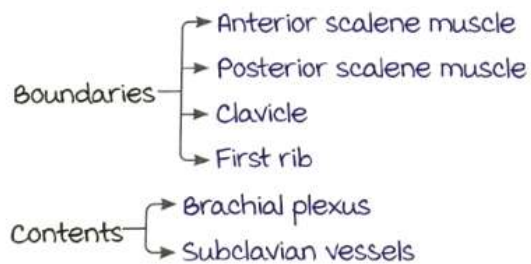


→ Belly of biceps muscle

Popeye sign

THORACIC OUTLET SYNDROME

Entrapment of neurovascular structures as they pass through the thoracic outlet.

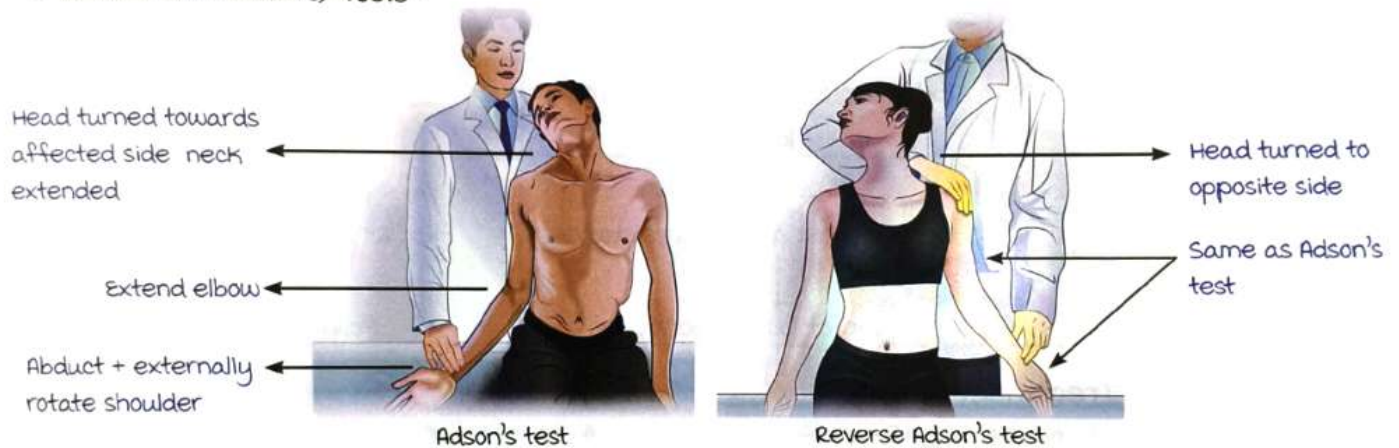
Thoracic outlet :**Causes of obstruction :**

- Abnormal positioning of neck.
- Lung tumor (upper lobe) causing compression.
- Cervical rib (Adventitious rib at C7).

----- Active space -----

Symptoms :

- Pain around shoulder, radiating to arm.
- Hand turns pale.
- Neurological Progresses to → vascular (Symptoms of claudication)
 - ↓
 - compression of brachial plexus at C8, T1
 - ↓
 - Humpke's palsy
- Risk of compression : Brachial plexus > Artery > Vein.

Evaluation :**1. Clinical (Provocative) tests :****Hyperabduction of shoulders****c. Wright's hyperabduction test :**

a. MRI : IOC.

Treatment :**Conservative :**

- Rest.
- Splint.
- Physiotherapy.

Surgical :

Excision of first rib/cervical rib/tumor (Lung).

- Abduct shoulder + Flex elbow.
- Open & close hands for 3 mins.

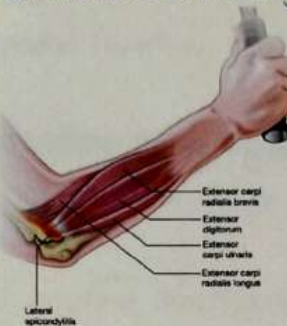
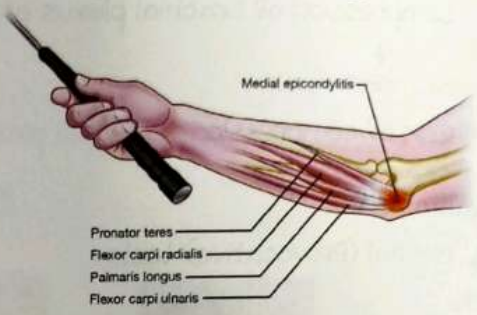
Reproduces symptoms of vascular claudication

**d. Roo's test :**

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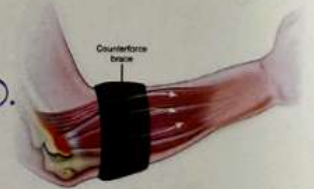
Conditions Around Elbow

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	Tennis elbow (m/c)	Golfer's elbow
Pathology	Lateral epicondylitis (Inflammation of common extensor origin) 	medial epicondylitis (Inflammation of common flexor origin) 
Earliest affected muscle	Extensor carpi radialis brevis > longus	Flexor carpi radialis
Evaluation	Cozen's test : Extend wrist against resistance ↓ Pain at lateral condyle	Reverse cozen's test : Flex wrist against resistance ↓ Pain at medial condyle

Treatment :

1. Activity modification.
2. NSAIDs.
3. Ortho-biologics (Helps in healing) :
 - a. Blood : ↑ healing and ↑ inflammation.
 - b. Platelet-rich plasma : Only ↑ healing (WBCs removed).
 - c. Stem cells from :
 - Adipose tissue.
 - Bone marrow.
 - Periosteum.
4. Counterforce brace (Prevents movement from reaching the origin of muscle)

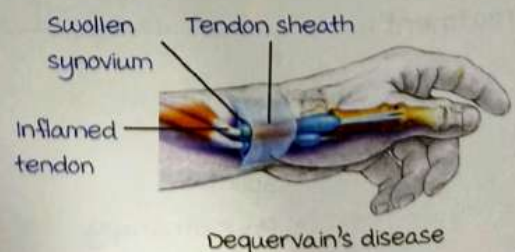
**Conditions Around Wrist & Hand**

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DeQuervain's disease :

stenosing tenosynovitis of { Abductor pollicis longus } At the first dorsal compartment of wrist
 { Extensor pollicis brevis }

etiology { Overuse,
 { Rheumatoid arthritis.



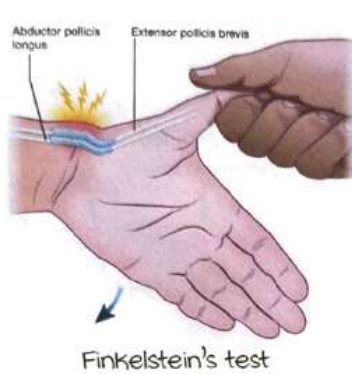
Dequervain's disease

Clinical features :

- F > m.
- Pain over radial styloid.

----- Active space -----

Evaluation :



Treatment :

- Rest.
- Immobilisation (Splints).
- Steroid injections.

Dupuytren's Contracture :

Abnormal cord like fibrosis of palmar fascia → Contracture of finger.



Dupuytren's contracture

Clinical features :

- m > F.
- Bilateral.
- m/c involved → Finger : Ring finger
Joint : metacarpophalangeal joint

Ectopic types (In other parts of the body)

Peyronie's
(Penis)



Ledderhose disease
(Soles of feet)



----- Active space -----

Treatment :

1. Conservative.
2. Collagenase injections.
3. Fasciectomy.

Trigger finger :

Pathophysiology :

Trauma → Stenosis of A-1 pulley

Obstruction of flexor tendon passing through it

Difficulty in opening the finger

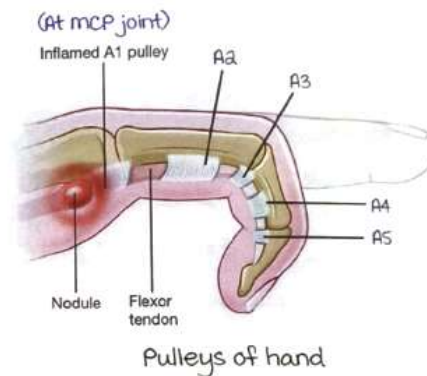
Finger suddenly opens on forceful extension

Etiology →

- Trauma (m/c)
- Rheumatoid arthritis

Clinical features :

- F > m.
- m/c involved →
 - Finger : Ring finger
 - Tendon : Flexor digitorum profundus (FDP)

**Gamekeeper's thumb / Skier's thumb**

Hyperabduction of thumb

Tear/avulsion of ulnar collateral ligament (UCL) of thumb.

Treatment :

- Thumb spica
- Surgical repair : If stener lesion (+)
(Adductor pollicis tendon trapped between the torn UCL)



Gamekeeper's thumb

Bowler's thumb :

Neuroma of ulnar digital nerve of thumb (D/t perineural fibrosis).

